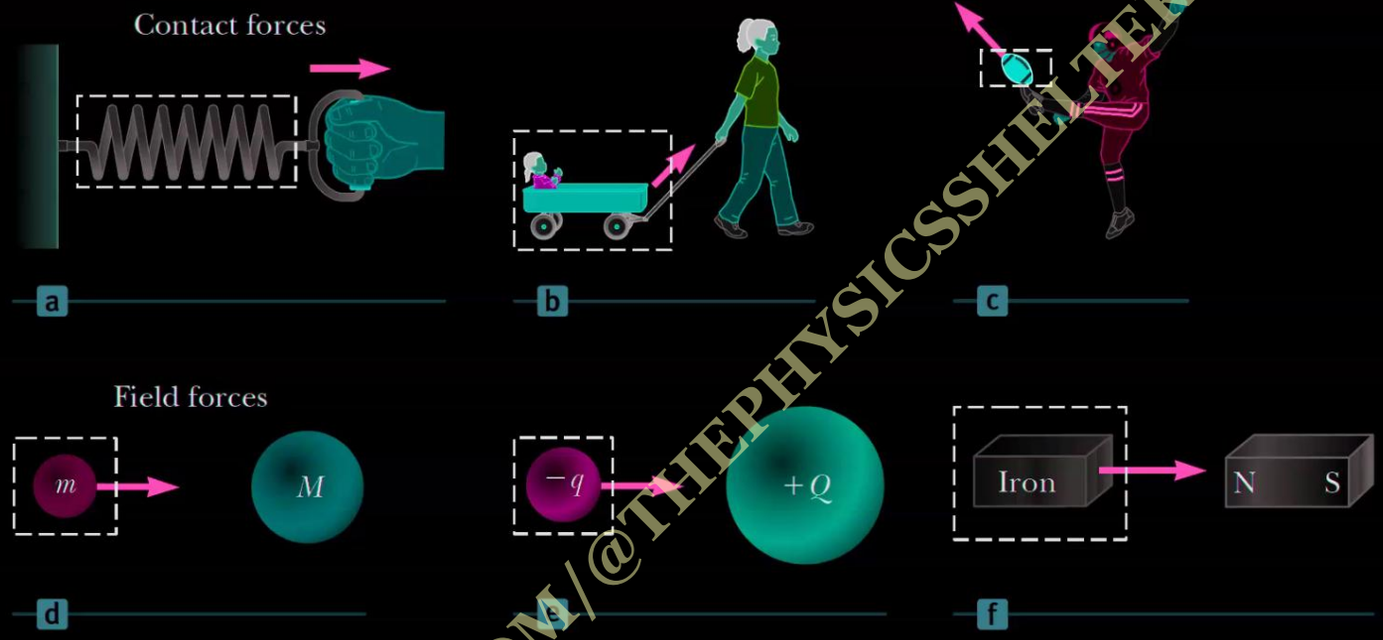
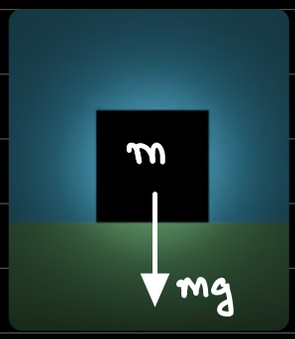
FORCE: A force is a push or a pull or a shear, or a combination. A force directed towards a surface is a push, a force directed away from a surface is pull, while a force directed parallel to a surface is called shear. The forces may be further classified as contact forces and non-contact forces. Non-contact forces such as gravitational force, magnetic forces and electrostatic forces do not require physical contact for action. Non-contact forces are also called field forces.

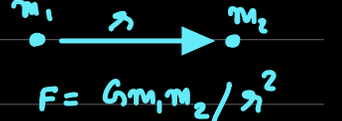


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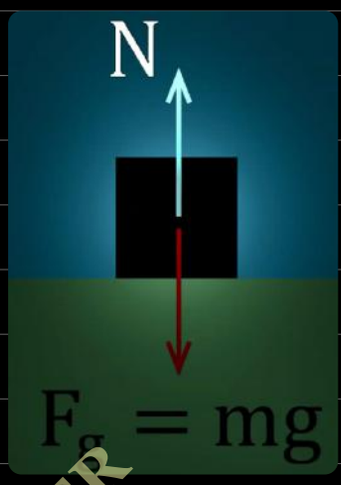
SOME TYPES OF FORCES:

GRAVITATIONAL FORCE: In the universe, every particle attracts every other particle, accordingly earth also attracts any object placed on or near it. For purpose of this chapter we will assume acceleration due to gravity as constant (However this is not actually true but an approximation). Thus if any object of mass m is placed on a surface, the gravitational force on it is taken as $m\vec{g}$ which is directed towards centre of earth.



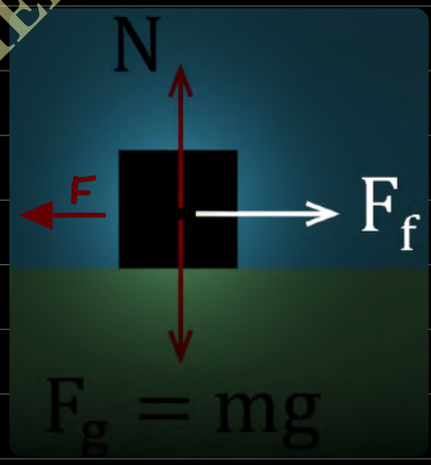


Normal Contact Force Between A Pair Of Surfaces:



Whenever two surfaces are in contact, they tend to push against each other due to very close distance electromagnetic interactions. This pushing force is called Normal reaction (or Normal contact force). It is a measure of how hard particles of contacting surfaces are pushing against each other.

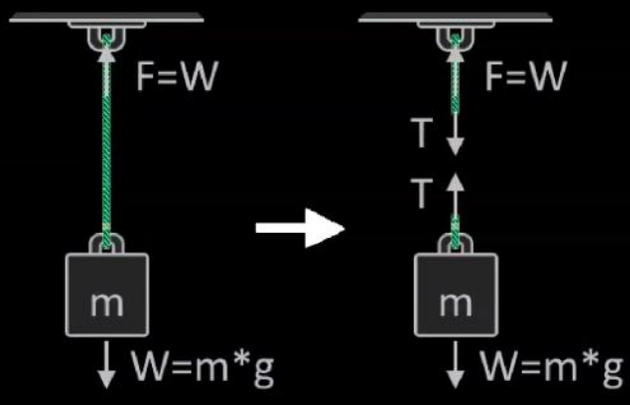
Friction: Whenever one surface is rubbing against another, or has a tendency to rub, a contact force acts in a direction opposite to this tendency. This contact force is called friction. Friction is also due to close contact electromagnetic interactions. (When two surfaces are close together they form bonds, which need to be broken for starting the motion).



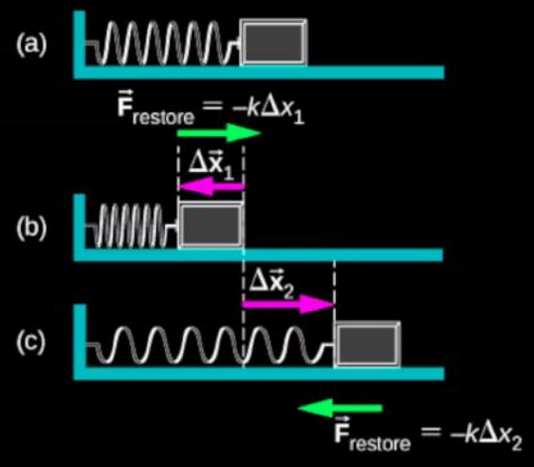
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String Tension:

Whenever an object is connected to a taut string, the string exerts a pulling force on the object. This is known as string tension force. It is always directed along the string and away from the object for a massless non rigid string. Not only is the tension force experienced by the connected body, but the particles of the string also pull each other with the string tension force.



Spring Force:



Whenever a block is connected to a deformed spring, the spring exerts a force on it which is proportional to the deformation in the spring. If the spring is compressed by a length x, it exerts a pushing force kx while if it is stretched by a distance x, it exerts a pulling force kx where k is the spring constant which depends on material and geometry of the spring.

Newton's Laws Of Motion:

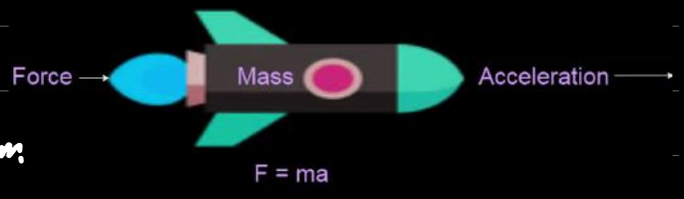
Newton's First Law: (Law of Inertia):

A body continues to be in a state of rest or uniform motion unless acted upon an external unbalanced force.

→ Newton's first law is given to define inertial frames. There can be frames in which the first law fails, such frames are called non inertial frames. For instance if I sit on a revolving chair, and start rotating on it, from my frame the whole universe has started rotating in the opposite direction without introduction of any additional force. Thus such a frame of observation is non-inertial. For analysis of motion in non-inertial frames, we have to introduce fictitious forces, such forces are called inertial forces (i.e the forces required to make non-inertial frames behave like inertial frames. Inertial forces are commonly called pseudo forces).

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Newton's Second Law Of Motion:



Net force acting on a particle is equal to rate of change of its momentum.
Corollary: Net external force on a system is equal to rate of change of linear momentum of the system.
This can be illustrated as follows:

$$\vec{F}_i = \frac{d}{dt}(\vec{p}_i)$$ (i: i^{th} particle, $\vec{p}$: momentum).

$$\Rightarrow \Sigma \vec{F}_i = \frac{d}{dt} \Sigma \vec{p}_i$$

$$\Rightarrow \Sigma \vec{F}_{i\,int} + \Sigma \vec{F}_{i\,ext} = \frac{d}{dt} \Sigma \vec{p}_i$$

Now since all internal forces are action reaction pairs by Newton's third law thus they cancel out on summation. Thus we are left with

$$\vec{F}_{ext} = \frac{d}{dt}\left[\vec{p}_{sys}\right] \quad (1)$$

Further if system consists of constant mass particles then $\vec{p}_i = m_i \vec{v}_i$.
$\Rightarrow \frac{d}{dt}\vec{p}_i = m_i \vec{a}_i$ thus our equation (1) becomes:

$$\boxed{\vec{F}_{ext} = \Sigma m_i \vec{a}_i} \quad (2)$$

→ Equation (2) is often more useful than applying Newton's second law to each particle because it helps us avoid unknown internal force variables in our equations. Thus using (2) we will often generate an easier system of equations for solving mechanical systems.

→ For a single particle, (2) reduces to $\vec{F} = m\vec{a}$

→ $\because \vec{F} = d\vec{p}/dt \Rightarrow \vec{F}\,dt = d\vec{p} \Rightarrow \int_0^T \vec{F}\,dt = \int_{\vec{p}_i}^{\vec{p}_f} d\vec{p}$

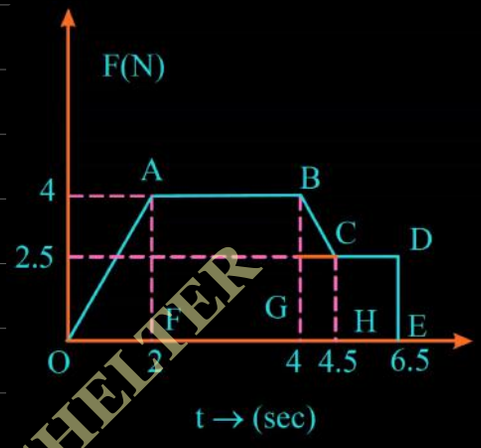
$\Rightarrow \vec{J} = \Delta\vec{p}$ where $\vec{J} = \int_0^T \vec{F}\,dt$ is called impulse of the

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force. Further if the force is acting only along a single axis, we can treat it as a signed scalar and then $\int F\,dt$ is simply area under the F-t graph.

→ $\vec{J} = \Delta\vec{p}$ or $\vec{J}_{ext} = \Delta\vec{p}_{sys}$ is called linear impulse momentum theorem.

EXAMPLE: A body of mass 2 Kg has an initial speed of $5\,ms^{-1}$ along positive x axis. A force acts on it along direction of its motion, whose variation with time is shown in the adjacent figure. Find the final speed of the body.



Solution: Total area under the graph = change in linear momentum.

∵ entire graph is above t axis, all areas are considered positive. Thus J = Net area = 18.625 Ns.

Now $J = \Delta p = m\Delta v = m(v-u) \Rightarrow v = \frac{J}{m} + u$

$\Rightarrow v = \frac{18.625}{2} + 5 = +14.25\ ms^{-1}$.

Think: What if the force were always along negative x axis?

EXAMPLE: The force on a bullet fired from a gun is $F = a - bt$ N The force reduces to zero just when the bullet leaves the Bannel of the Gun. (a) What is the total impulse on the bullet (b) What is the velocity of the bullet if its mass is m. a and b are positive constants.

Solution: $a - bT = 0 \Rightarrow T = a/b$ (T: Time of exit).

Now $J = \int F\,dt = \int_0^{a/b} (a-bt)\,dt = a^2/2b$

Now by linear Impulse Momentum theorem: $\vec{J} = \Delta\vec{p}$ thus $v = J/m$

i.e $v = a^2/2mb$ (v: exit velocity).

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NEWTON'S THIRD LAW OF MOTION:

If an object A exerts a force on object B "action", then object B exerts a force on object A "reaction". These two forces have same magnitudes but opposite directions. These two forces act on different objects and act along the same line. (All real forces have a pair).

→ Just because two forces are equal and opposite does not necessarily make them action and reaction. The two forces must be acting along same line and on different bodies. For clarity study the following illustration carefully.

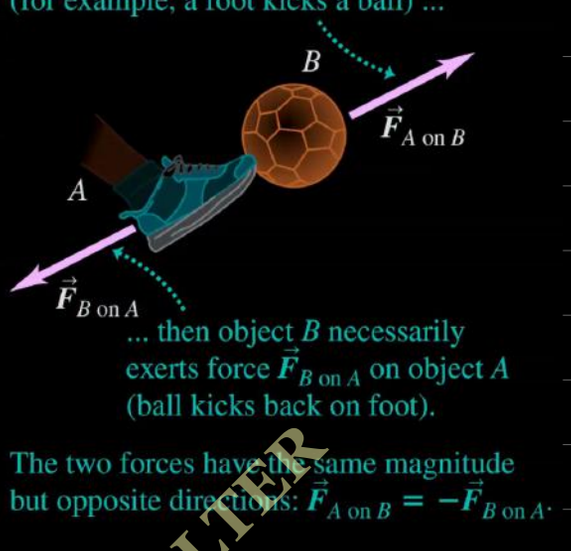
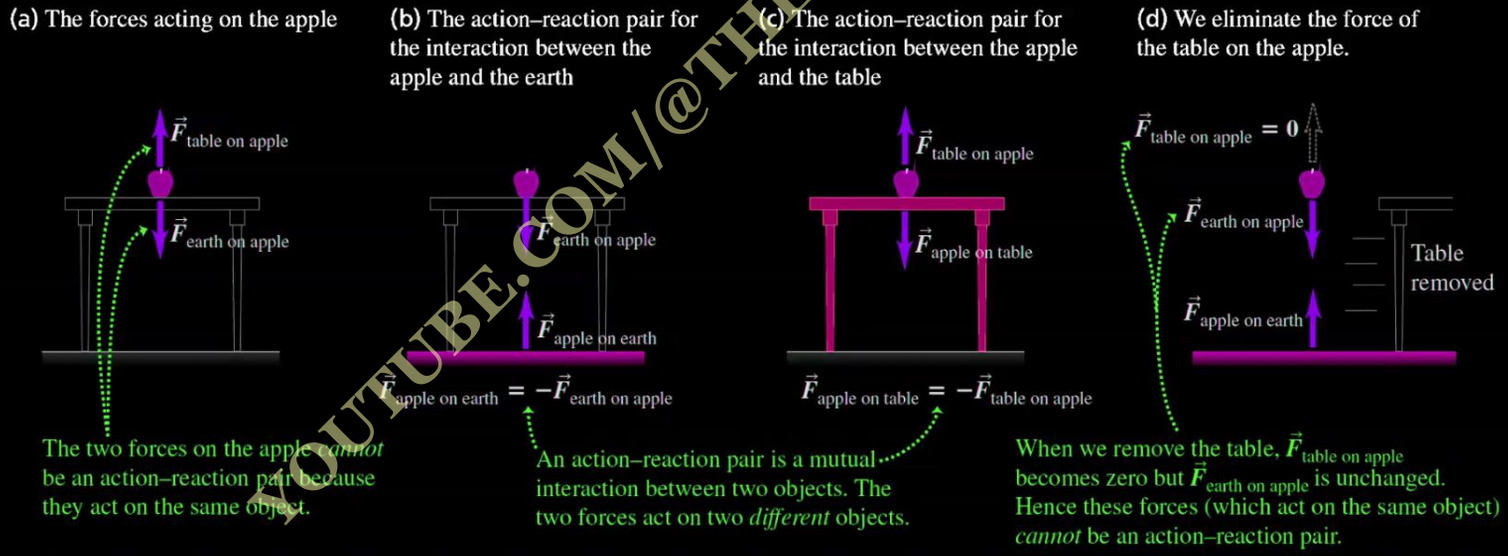


ILLUSTRATION:

Identifying action–reaction pairs.

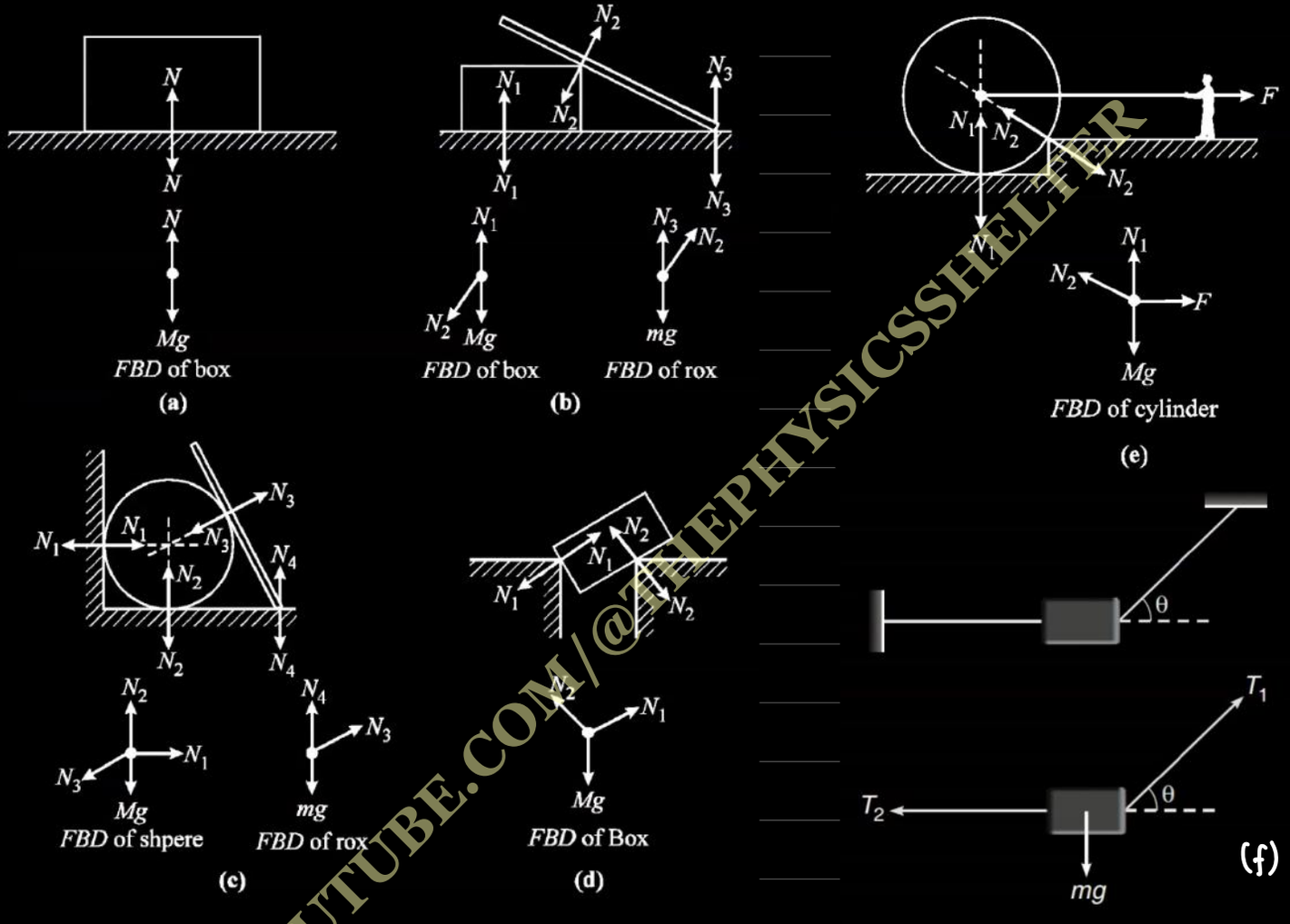


→ Here Normal reaction on apple and its weight are equal and opposite however they are not action and reaction because they are acting on the same body.

FREE BODY DIAGRAMS: A diagram representing all the forces acting on a body is called its free body diagram. For drawing the free body diagram, all the ropes and surfaces touching a body are removed and replaced by the forces offered by them. In addition,

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remember to show the non-contact forces also. (Apart from forces if couples are acting on bodies they must also be shown in the FBD, however for this chapter we will be confined to forces only. A couple is a force system where vector sum of forces is zero but still it can produce a rotational effect). Some authors chose to show bodies as particles while drawing the FBD however it can be done only for non rotating bodies and is not necessary.



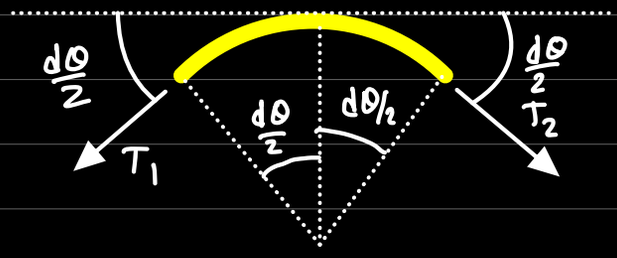
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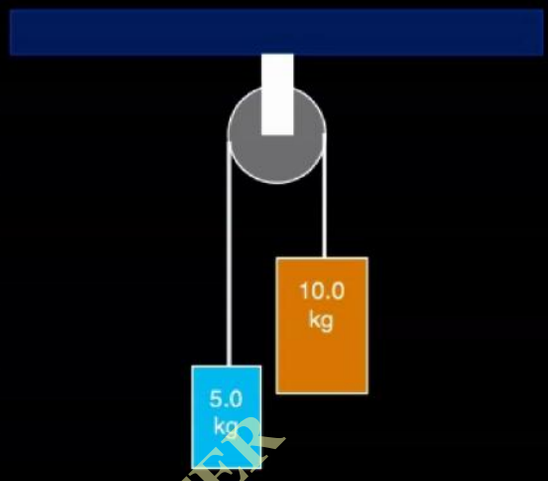
→ Note that whenever there is a point and surface contact, normal is always defined by the ⊥ to the surface (illustrations b, c, d and e).

FORCE ON A MASSLESS OBJECT:

While any material body has some mass, for the purpose of framing problems, we do come across massless bodies. Now since $\vec{F} = m\vec{a}$, for a massless body net force must always be zero.

Tension In A Massless Rope Passing Over Frictionless Surfaces:

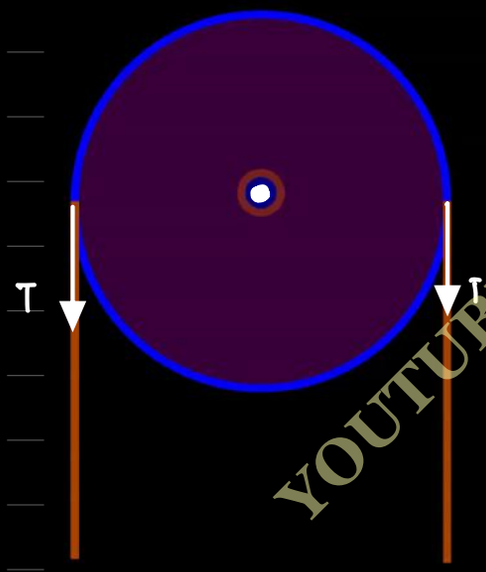




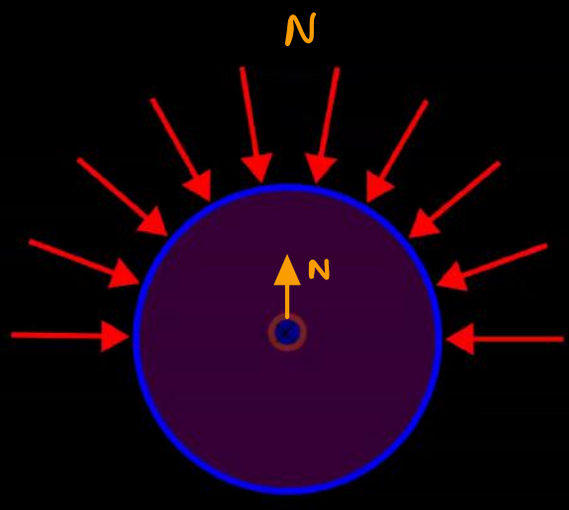
Consider the FBD of a small section of a massless rope as shown in the figure above. Here by horizontal force balance, $T_1 \cos(d\theta/2) = T_2 \cos(d\theta/2)$, thus $T_1 = T_2$

Thus whenever we have a massless string passing over a frictionless surface, the tension throughout the string is same.

Free Body Diagram Of Massless Pulley:







When a rope passes over a frictionless and massless pulley, it exerts a normal reaction on the pulley groove and in turn experiences a distributed normal reaction as shown above. Now Since net force on a massless object is zero ∴ The distributed Normal reaction resultant is exactly equal and opposite to the tension forces. Thus instead of showing the detailed FBD of the pulley with distributed normal reactions, we can draw an effective equivalent as shown

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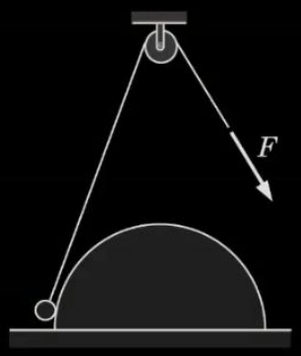
in the first figure, where we just show the two rope tensions acting on the pulley.

EQUILIBRIUM: A particle is said to be in equilibrium if net resultant of all forces acting on it is zero. (For extended bodies, net torque of all forces must also be zero, this will be proved in the chapter on rotational mechanics).

→ In case of equilibrium under action of three forces, the three forces must form a closed triangle (according to polygon law of addition) or alternatively they should satisfy Lami's theorem as will be discussed in a coming section.

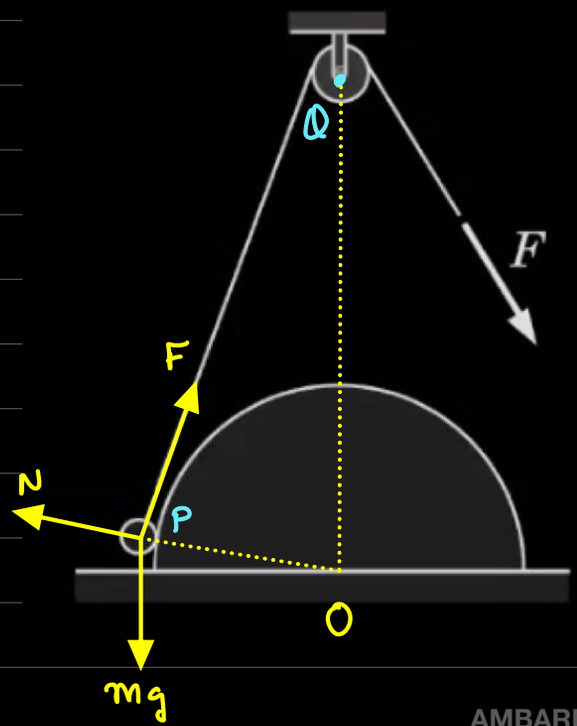
EXAMPLE: PATHFINDER MCQ-7:

A small metal ball is being pulled gradually on a fixed frictionless hemisphere as shown in the figure. Radii of the ball and that of the pulley are much smaller than that of the hemisphere. As the ball slides from the bottom to a position close to the top of the hemisphere, how do the magnitudes of pulling force F and contact force N between the ball and the hemisphere change?



(a) F increases and N decreases.
(b) F decreases and N increases.
(c) F decreases and N remains unchanged.
(d) F remains unchanged and N decreases.

Solution: We observe here that the three forces acting on the ball are $\vec{F}$ which is // to side $\vec{PQ}$ of ΔOPQ, $m\vec{g}$ which is parallel to side $\vec{QO}$ and $\vec{N}$ which is // to side $\vec{OP}$, thus sides of the closed triangle formed by forces $\vec{F}$, $m\vec{g}$ and $\vec{N}$ must be parallel to sides of ΔOPQ and hence the triangle will be similar to ΔOPQ.



Thus:

$$\frac{mg}{OQ} = \frac{F}{PQ} = \frac{N}{OP}$$

Here OP and OQ remain constant as the

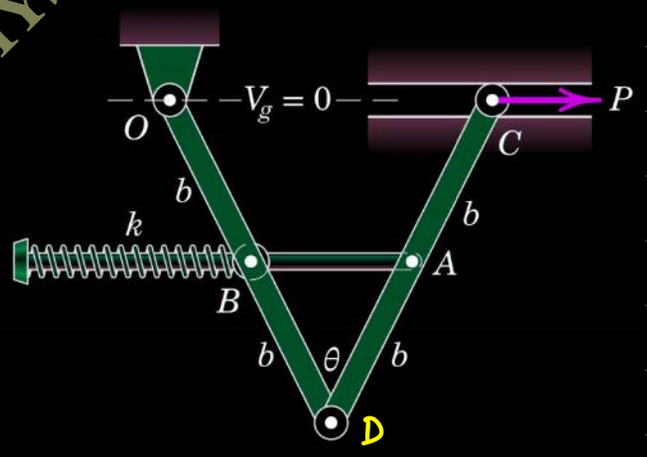
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ball is pulled, however PQ reduces in length hence, N remains constant and F reduces. Correct option (C)

A MASSLESS PIVOTED ROD: Whenever we have massless rod pivoted at its ends and experiencing some load at end points, if the force is tensile, then such a rod behaves just like a string while if it is in compression, compressive forces at both ends are equal and opposite and along the rod. (If instead of pivots we have welds then, the force can have a component perpendicular to the rod as well, for if there are other forces tending to bend the rod, then also the force may not be along the rod but can also have component ⊥ to rod).

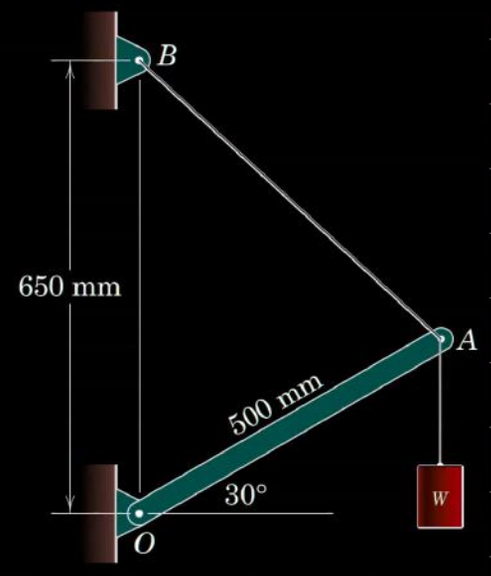


→ For instance, in the adjacent figure, the tension in rod AB is along AB itself, however in members OD and DC, the forces at end points will in general not be along the rods. (∵ at A and B also they are experiencing some forces).



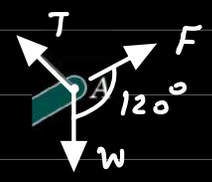
EXAMPLE: In the adjacent rod, a weight W hangs from the pivot at point A which is also connected to a string and the rod is massless. (a) Find the compressive force in the rod (b) Find tension in the string AB.

Solution: Considering FBD of the pivot pin at A, it experiences a tension along AB, another tension W along vertical and a pushing force along rod (say F). We see that these three forces are parallel to the sides of ΔOBA. Thus using similar triangles:



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$$\frac{F}{500} = \frac{W}{650} \Rightarrow F = \frac{500}{650} \times W = \frac{10W}{13} \text{ Ans.}$$



Now T is just balancing resultant of F & W, thus:

$$T = \sqrt{F^2 + W^2 + 2FW\cos 120°} = \frac{\sqrt{139}}{13} W \text{ Ans.}$$

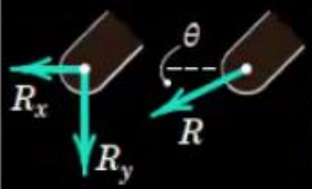
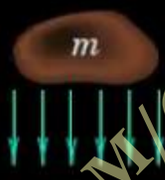
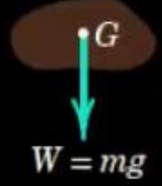
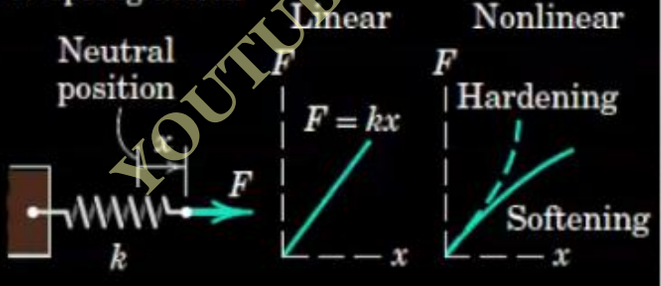
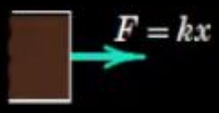
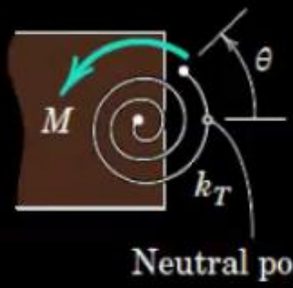
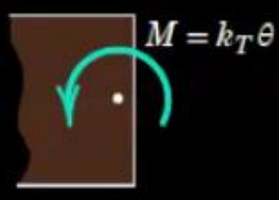
Modeling The Action Of Forces In Two Dimensions:

MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS

Type of Contact and Force Origin	Action on Body to Be Isolated
1. Flexible cable, belt, chain, or rope Weight of cable negligible Weight of cable not negligible	Force exerted by a flexible cable is always a tension away from the body in the direction of the cable.
2. Smooth surfaces	Contact force is compressive and is normal to the surface.
3. Rough surfaces	Rough surfaces are capable of supporting a tangential component F (frictional force) as well as a normal component N of the resultant contact force R.
4. Roller support	Roller, rocker, or ball support transmits a compressive force normal to the supporting surface.
5. Freely sliding guide	Collar or slider free to move along smooth guides; can support force normal to guide only.

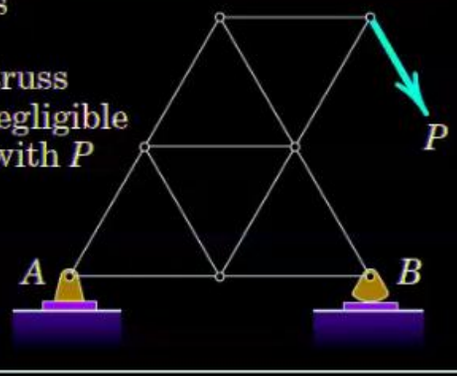
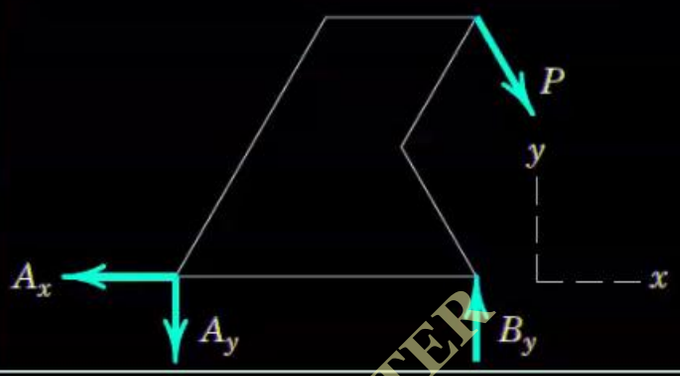
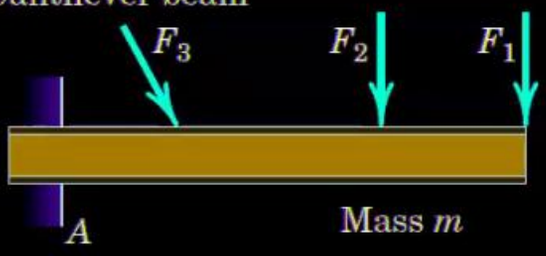
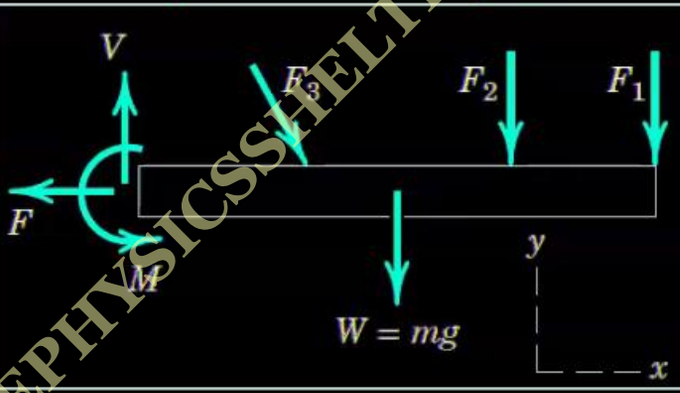
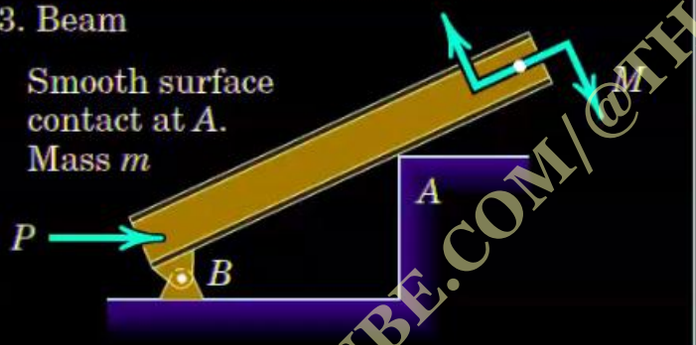
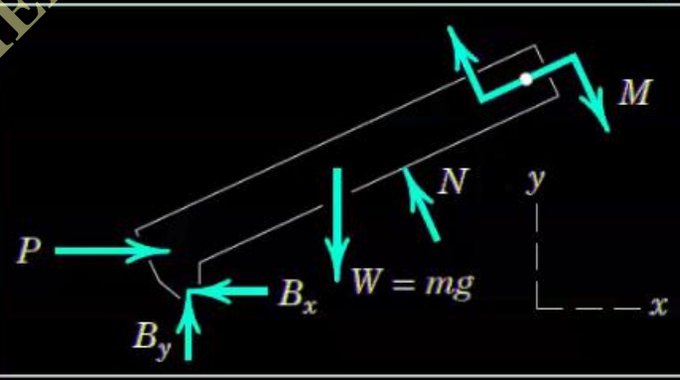
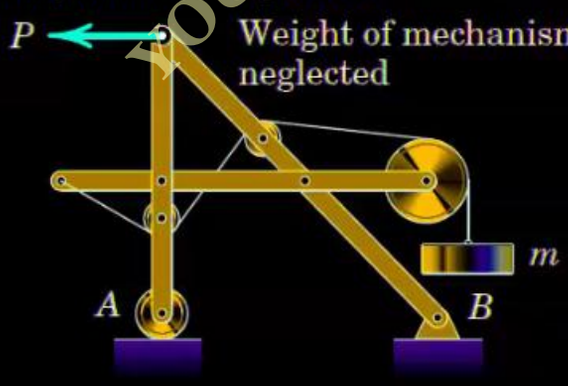
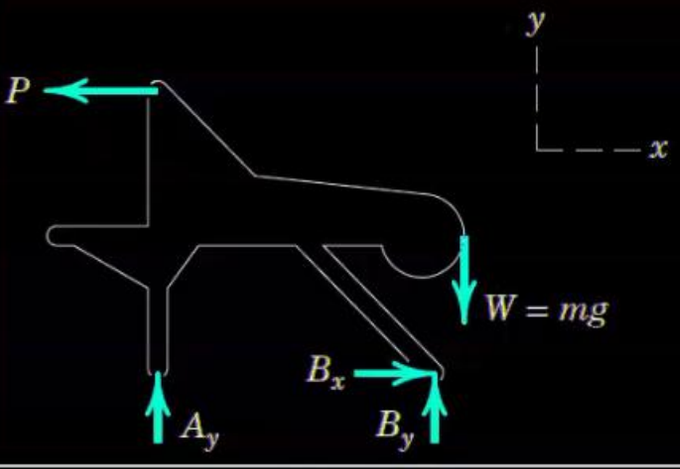
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MODELING THE ACTION OF FORCES IN TWO-DIMENSIONAL ANALYSIS (*cont.*)

Type of Contact and Force Origin	Action on Body to Be Isolated
6. Pin connection 	Pin free to turn  Pin not free to turn  A freely hinged pin connection is capable of supporting a force in any direction in the plane normal to the pin axis. We may either show two components R_x and R_y or a magnitude R and direction θ. A pin not free to turn also supports a couple M.
7. Built-in or fixed support  	 A built-in or fixed support is capable of supporting an axial force F, a transverse force V (shear force), and a couple M (bending moment) to prevent rotation.
8. Gravitational attraction  	 The resultant of gravitational attraction on all elements of a body of mass m is the weight $W = mg$ and acts toward the center of the earth through the center of gravity G.
9. Spring action  	 Spring force is tensile if the spring is stretched and compressive if compressed. For a linearly elastic spring the stiffness k is the force required to deform the spring a unit distance.
10. Torsional spring action  	 For a linear torsional spring, the applied moment M is proportional to the angular deflection θ from the neutral position. The stiffness k_T is the moment required to deform the spring one radian.

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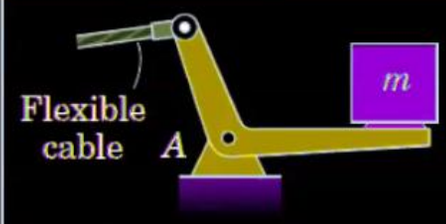
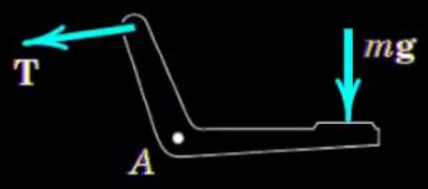
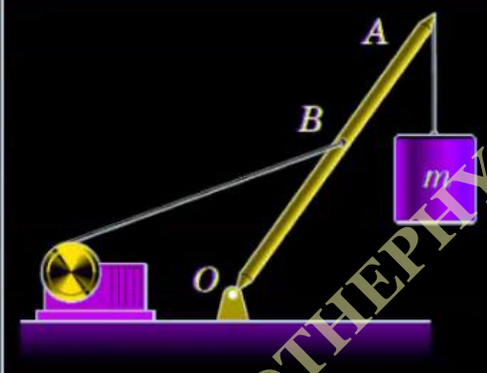
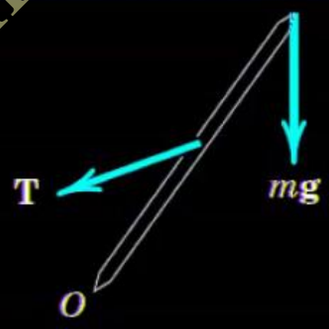
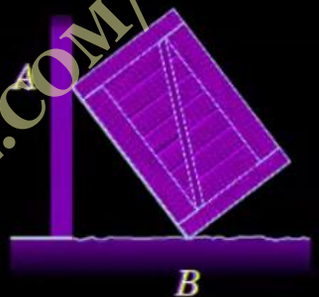
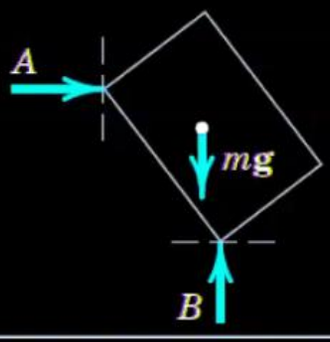
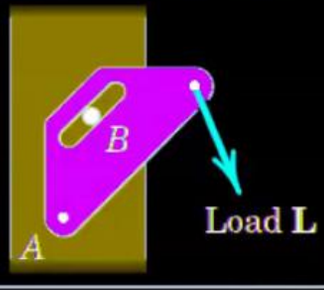
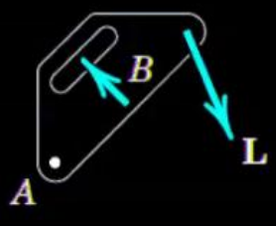
Some More Examples Of FBD:

SAMPLE FREE-BODY DIAGRAMS	
Mechanical System	Free-Body Diagram of Isolated Body
1. Plane truss Weight of truss assumed negligible compared with P 	
2. Cantilever beam 	
3. Beam Smooth surface contact at A. Mass m 	
4. Rigid system of interconnected bodies analyzed as a single unit 	

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→ If certain bodies / parts in mechanical systems are not moving wrt each other we can always make their combined FBD. For example consider the cases 1 and 4 above.

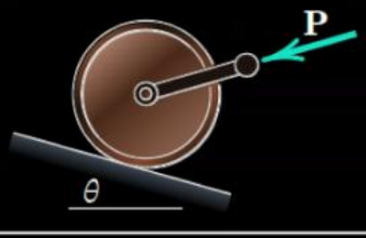
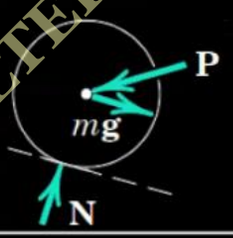
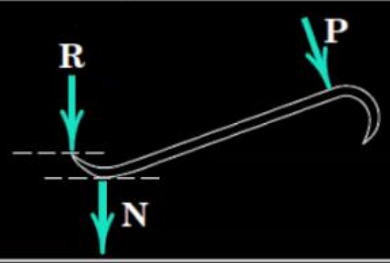
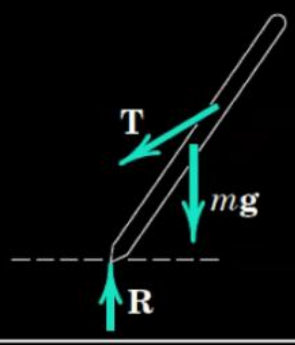
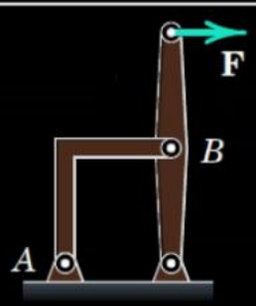
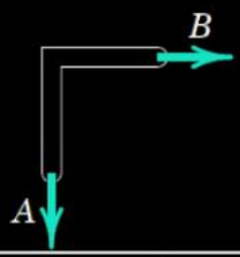
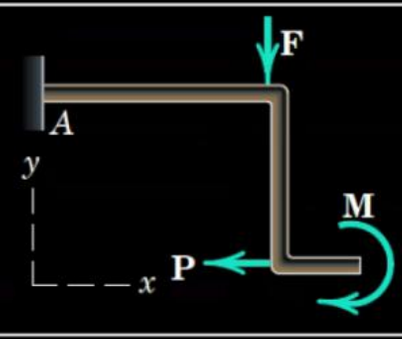
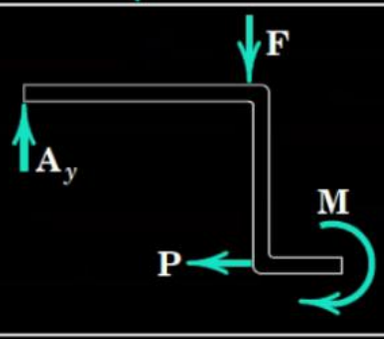
EXAMPLE: Identify the missing Forces / Torques in the incomplete FBDs:

	Body	Incomplete FBD
1. Bell crank supporting mass m with pin support at A.		
2. Control lever applying torque to shaft at O.		
3. Boom OA, of negligible mass compared with mass m. Boom hinged at O and supported by hoisting cable at B.		
4. Uniform crate of mass m leaning against smooth vertical wall and supported on a rough horizontal surface.		
5. Loaded bracket supported by pin connection at A and fixed pin in smooth slot at B.		

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Solution:
1. Pin reactions at A missing
2. Reaction moment at O missing
3. Pin reactions at O missing
4. Friction at B missing
5. Pin reactions at A missing.

Think! What is wrong / missing in following FBDs:

	Body	Wrong or Incomplete FBD
1. Lawn roller of mass m being pushed up incline θ.		
2. Prybar lifting body A having smooth horizontal surface. Bar rests on horizontal rough surface.		
3. Uniform pole of mass m being hoisted into position by winch. Horizontal supporting surface notched to prevent slipping of pole.		
4. Supporting angle bracket for frame; pin joints.		
5. Bent rod welded to support at A and subjected to two forces and couple.		

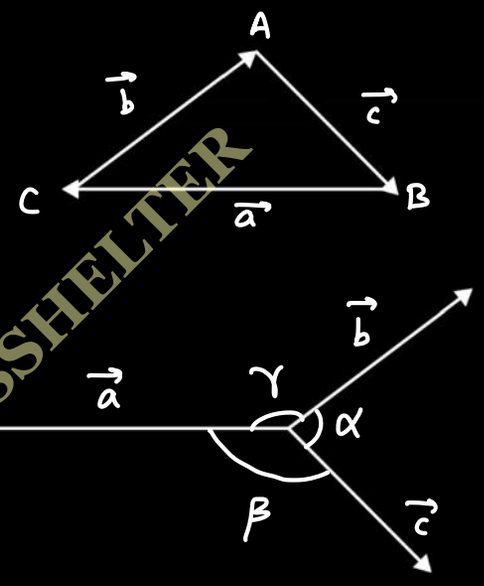
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Solution:
1. Direction of mg is wrong, friction is missing
2. Wrong direction of normal reaction from ground, friction missing
3. Horizontal reaction at the notch missing.
4. Horizontal pin reaction at A and vertical at B missing
5. Reaction moment at A missing.

LAMI'S THEOREM:

If a particle is in equilibrium under action of three forces $\vec{a}, \vec{b}$ and $\vec{c}$ as shown, then if the forces are drawn in coinitial configuration, then:

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma} \quad \text{①}$$



Proof: Consider ΔABC, by sine law:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \text{②}$$

$$\Rightarrow \frac{a}{\sin(180°-A)} = \frac{b}{\sin(180°-B)} = \frac{c}{\sin(180°-C)} \quad \text{③}$$

Now recognize that angle between $\vec{b}$ and $\vec{c}$ is $180°-A = \alpha$ similarly visualize for other pairs thus Equation ③ gives:

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma} \quad \text{QED.}$$

SECTIONAL FBD: Just as for bodies with no relative motion we can draw combined FBD, we can also visualise a single body to be made of multiple sections and can draw FBD for each section separately.

EXAMPLE: A chain of weight W hangs from a ceiling as shown. Find the tension in the chain as a function of distance x from the bottom.



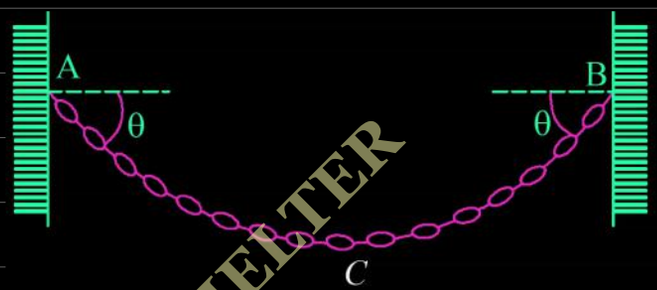
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Solution: Considering the sectional FBD of a length x of the rope, $T = \frac{Wx}{\ell}$ Ans.

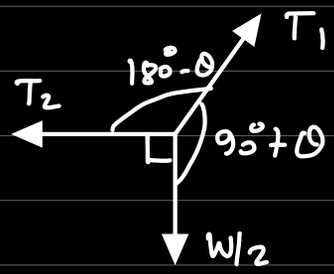
(x/ℓ is fractional length and therefore also weight fraction, thus weight of the section is Wx/ℓ).

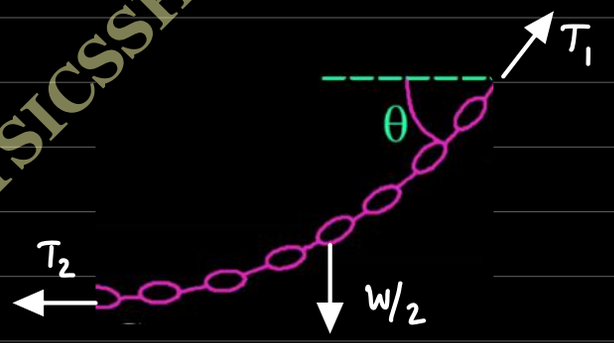


EXAMPLE: A chain of weight W hangs from two walls as shown in the figure. Find the tension at the supports and the bottommost point (C).



Solution: Method I: Lami's theorem:





$$\frac{T_1}{\sin 90^\circ} = \frac{T_2}{\sin(90^\circ+\theta)} = \frac{W/2}{\sin(180^\circ-\theta)} \Rightarrow T_1 = \frac{W}{2}\operatorname{cosec}\theta \quad T_2 = \frac{W}{2}\cot\theta \text{ Ans.}$$

Method 2: Force Balance:

$T_1\cos\theta = T_2$ (1) $(\Sigma F_x = 0)$

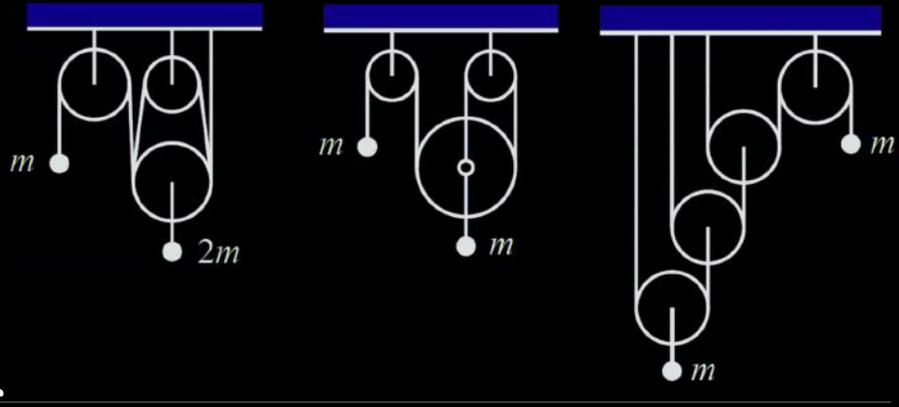
$T_1\sin\theta = W/2$ (2) $(\Sigma F_y = 0)$

$$\Rightarrow \frac{2T_2}{W} = \cot\theta \Rightarrow T_2 = \frac{W}{2}\cot\theta$$

Now $T_1 = \frac{T_2}{\cos\theta} = \frac{W}{2}\operatorname{cosec}\theta$ Ans.

ATWOOD MACHINES:

Any system comprising of masses strings and pulleys can be called an Atwood Machine. (Named After George Atwood, a tutor at Cambridge University). The figure shows some Atwood Machine

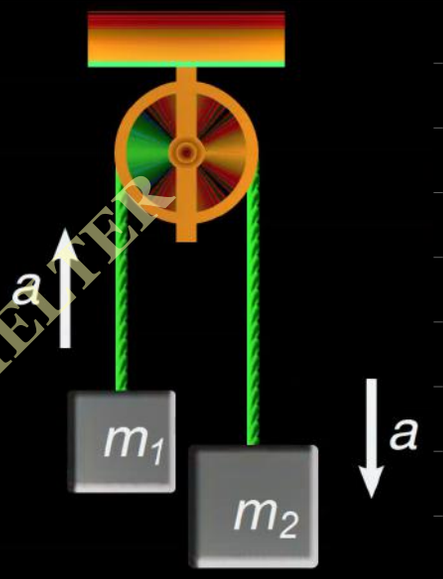


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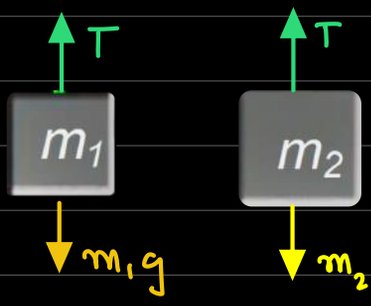
→ General analysis of Atwood Machines required some understanding of constraint relations which we will study in a later section. For now we will be studying a very simple atwood machine.

EXAMPLE: SIMPLE ATWOOD MACHINE:

Consider the atwood machine shown in adjacent figure ($m_2 > m_1$). Find (a) Acceleration a of the system (b) Tension in the rope (c) Tension in the link between ceiling and the pulley pin.



Solution:



Consider the FBDs of the blocks above. Now applying second law of motion:

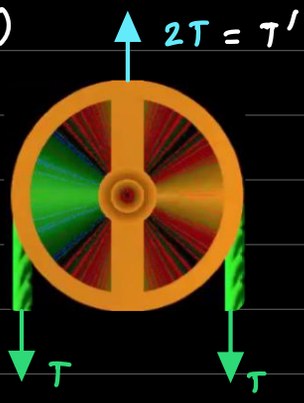
$$m_2 g - T = m_2 a \quad (1)$$

$$T - m_1 g = m_1 a \quad (2)$$

Now solving (1) & (2):

$$T = \frac{2 m_1 m_2}{m_1 + m_2} g \qquad a = \frac{(m_2 - m_1) g}{(m_2 + m_1)} \text{ ans.}$$

Now consider the FBD of the pulley + pin. Since net force on massless object must be zero, by force balance, we readily get the tension in link connecting ceiling to the pulley pin as $2T$ thus



$$\boxed{T' = \left(\frac{4 m_1 m_2}{m_1 + m_2}\right) g}$$

→ Thus for the ceiling, the system is as good as single block of mass:

$$\boxed{m_{eff} = 4 m_1 m_2 / (m_1 + m_2)}$$

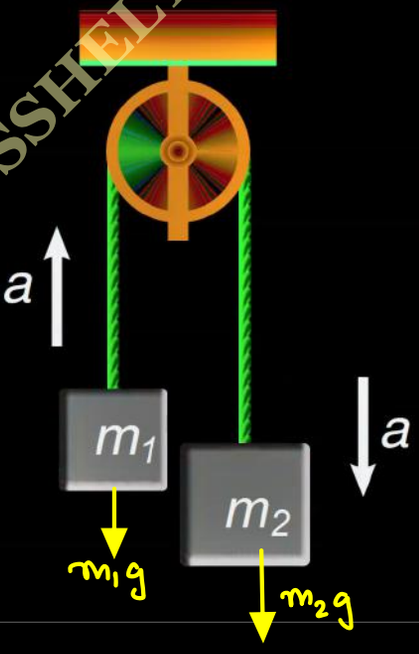
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→ This simplification must be memorized by heart. Many times, simple Atwood machine appears as a subsystem of complicated Atwood Machines and this simple substitution greatly simplifies the problem.

→ For simple linear chains where each element is moving along the chain with same magnitude of acceleration, acceleration can be simply found by considering the pulling force of external forces on the chain and then net acceleration is simply:

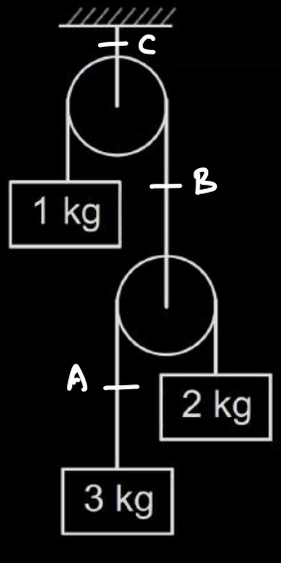
a = Net Pulling Force / Net mass of the chain.

→ For instance for the shown system, acceleration is considered downward positive on the right block and accordingly upward positive on the left block. Thus, net pulling force becomes: $F_p = m_2 g - m_1 g$ and net mass of chain is $(m_1 + m_2)$. Thus acceleration is simply $\frac{(m_2 - m_1)}{m_2 + m_1} g$



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EXAMPLE: For the Atwood Machine shown in the adjacent figure find the accelerations of the blocks and tensions in the strings A, B and C. (Pulleys and strings are ideal).



Solution: Simplifying 2-3 Atwood: $m_{23} = \frac{4 \times 2 \times 3}{2+3} = \frac{24}{5}$

Now further simplifying 23 combination with 1 Kg.

$m_{123} = \frac{4 \times 1 \times 24/5}{1 + 24/5} = \frac{96}{29}$ Kg.

Thus $T_C = \frac{960}{29}$ N, $T_B = \frac{T_C}{2} = \frac{480}{29}$ N, $T_A = \frac{T_B}{2} = \frac{240}{29}$ N

(∵ Net force on massless pulley is zero).

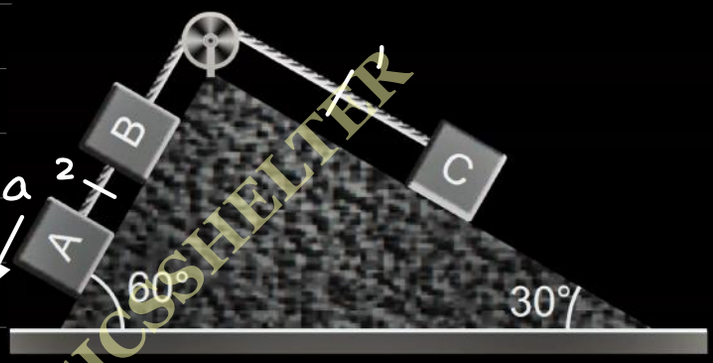
Now $a_1 = (T_B - 10)/1 = 6.55\ m/s^2$ upwards

$$a_3 = (30 - T_A)/3 = \left(30 - \frac{240}{29}\right)/3 = 7.24\ ms^{-2} \quad \text{(downwards)}$$

$$a_2 = (20 - T_A)/2 = (20 - 240/29)/2 = 5.86\ ms^{-2} \quad \text{(downwards)}.$$

(If any acceleration comes out to be negative, it means it is accelerated opposite to the assumed direction).

Example: In the adjacent figure masses of blocks A, B and C are 1Kg, 3 Kg and 2 Kg respectively. The pulley and strings are ideal and the wedge frictionless. Find the acceleration of the system and the tensions in each string.



Solution: With sense of acceleration as shown, pulling force along the chain becomes, $(1+3)g\sin 60° - 2g\sin 30° = 24.6N$, Total mass: $1+2+3 = 6Kg$

$\Rightarrow a = 24.6/6 = 4.1\ ms^{-2}$ Ans.

Now for block C:

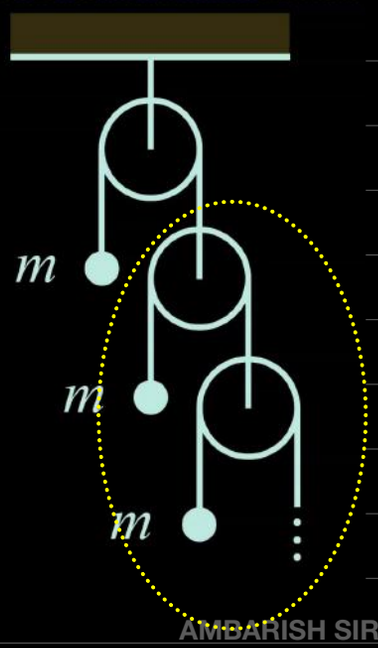
$$T_1 - 2g\sin 30° = 2 \times 4.1 \Rightarrow T_1 = 18.2N$$

For block A:

$$1g\sin 60° - T_2 = 1 \times 4.1 \Rightarrow T_2 = 4.6N \text{ Ans.}$$

Example: Infinite Atwood Machine:

Consider the infinite Atwood Machine shown in the adjacent figure. (a) What is the effective mass of the circled portion (b) What is the acceleration of the topmost mass.



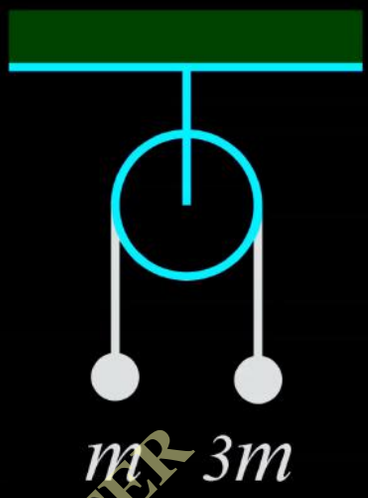
Solution: Let M be the effective mass of the circled portion. Now since the system in the dotted boundary is infinite, adding one more unit to the system should not make any difference to the effective mass. Thus:

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$$M = \frac{4mM}{M+m} \quad (1) \quad (\because \ m_{HF} = 4m_1m_2/(m_1+m_2))$$

Solving (1) for M, $M = 3m$.

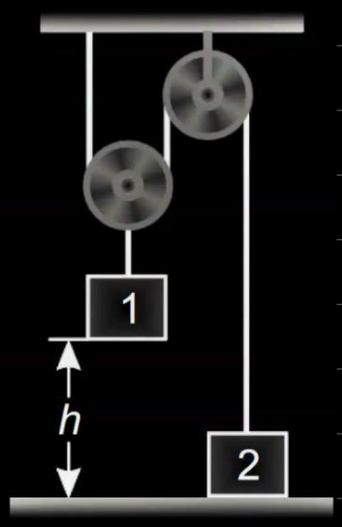
Thus the system reduces to one shown in adjacent figure. Now using pulling force method, $a = \left(\frac{3m-m}{3m+m}\right)g = g/2$ upwards.



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EXAMPLE: IRODOV 1.76:

In the arrangement shown in Fig. 1.19 the mass of body 1 is $\eta = 4.0$ times as great as that of body 2. The height $h = 20$ cm. The masses of the pulleys and the threads, as well as the friction, are negligible. At a certain moment body 2 is released and the arrangement set in motion. What is the maximum height that body 2 will go up to?

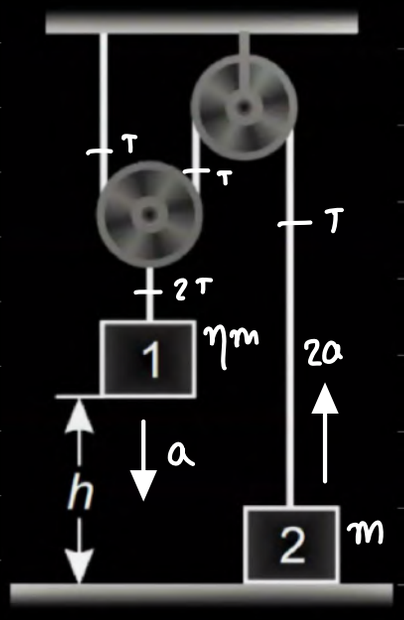


Solution: Since total length of the string is constant, by logic we can figure out that if body 2 goes up by 1 unit, body 1 will descend by 1/2 unit. This gives us the ratios of accelerations and velocities of the bodies. Once block 1 hits the ground, block 2 will continue moving up due to inertia. Now Writing LOM equations:

$$\eta mg - 2T = \eta ma \quad (1)$$

$$T - mg = 2ma \quad (2)$$

Solving, we get $a = g\left(\frac{\eta-2}{\eta+4}\right)$



Now time for touching the ground, $\tau = \sqrt{2h/a} = \sqrt{\frac{2h}{g}\left(\frac{\eta+4}{\eta-2}\right)}$

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By this time, block 2 has risen through a height $2h$ and has a velocity given by $v^2 = 2(2a) \times 2h = 8ah$ ($\because$ like $v^2 = 2gh$).
now further height travelled $h' = v^2/2g = \frac{8ah}{2g} = \frac{4a}{g}h$

$\Rightarrow h' = \frac{4(\eta-2)}{(\eta+4)}h$. Thus total height gained: $2h + h' = \frac{6\eta}{4+\eta}h$ Ans

EXAMPLE: IRODOV 1.74:

In the arrangement shown in Fig. 1.17. the mass of the rod M exceeds the mass m of the ball. The ball has an opening permitting it to slide along the thread with some friction. The mass of the pulley and the friction in its axle are negligible. At the initial moment the ball was located opposite the lower end of the rod. When set free, both bodies began moving with constant accelerations. Find the friction force between the ball and the thread if t seconds after the beginning of motion the ball got opposite the upper end of the rod. The rod length equals l.

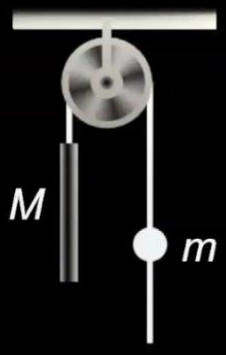


Fig. 1.17

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Solution: Let f be the friction force applied by the thread on the ball, then this must also be the tension in the thread because the thread is massless and a massless object must have a net zero force. This can be understood more clearly by considering FBD of portion of the string within the bead. The portion below the bead is hanging loose without any weight $\therefore$ has a zero tension. Thus $T = f$. ($\because$ massless thread).



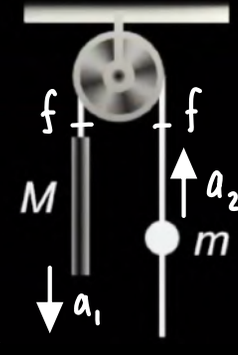
Now considering FBD of the rod and the bead:

$Mg - f = Ma_1$ ①

$f - mg = ma_2$ ②

Also relative acceleration $= a_1 + a_2$ and relative distance to be covered is l thus using relative kinematics:

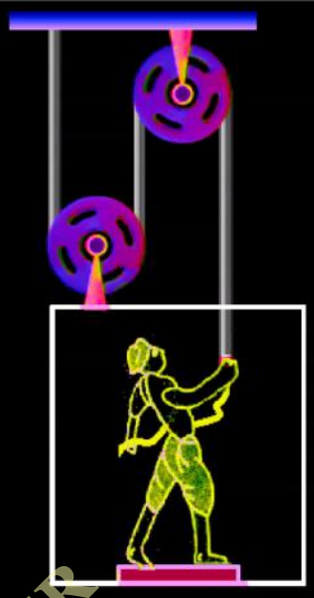
$\frac{1}{2}(a_1 + a_2)t^2 = l$ ③ Eliminating a_1 & a_2 and solving for f:



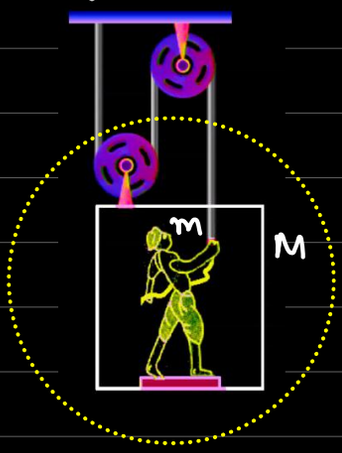
$f = 2Mml/(M-m)t^2$ Ans.

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EXAMPLE: A 70 Kg man is standing in a 50 Kg lift and pulling the string to keep the system stationary. Find (a) The force with which man pulls the string (b) Weight registered by weighing machine (c) Weight registered by weighing machine if the man doubles the pulling force.



Solution: If man pulls the string by a force T then by Newton's third law, string also pulls the man by a force T. We could do this problem by making individual FBD, however we prefer the "wizard" way over the "muggle" way. We will first use the combined and sectional FBD.



Imagine the dotted circle as a black box whose combined weight is $(m+M)g = 1200\,N$ and we see three threads coming out of it balancing the system. (Recall that whenever things are not moving relative to each other we can always make a combined FBD. Visualising in this manner, we get the FBD as shown in the adjacent figure.



Thus, $3T = (M+m)g \Rightarrow T = (M+m)g/3 = 400\,N$ Ans.

→ Weighing machine simply measures the Normal reaction exerted on it.

Thus now considering the FBD of the man alone, $T + N = mg \Rightarrow N = mg - T$

$\Rightarrow N = 700 - 400 = 300\,N$, thus weight recorded by weighing machine is just 300 N (or equivalently, man weighs 30 kg in this configuration).



(c) Now if man doubles the pulling force, i.e $T = 800\,N$, once again we go for the "wizard" way, combined / sectional FBD and then use the equation for combined systems as $\Sigma \vec{F}_{ext} = \Sigma m_i \vec{a}_i$.

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Thus: $2400 - 1200 = 70a + 50a = 120a$

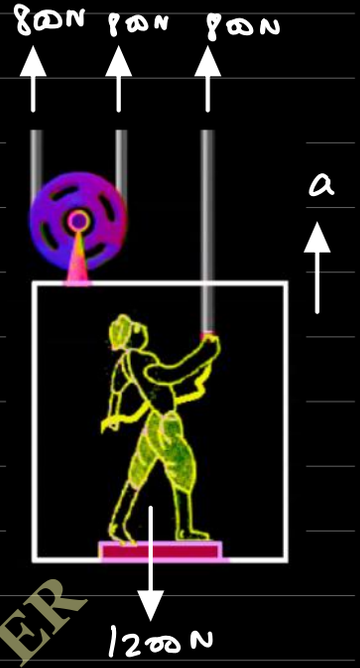
$\Rightarrow a = 10\ m/s^2$. Now considering FBD of the man alone:

$800 + N - 700 = 70 \times 10$

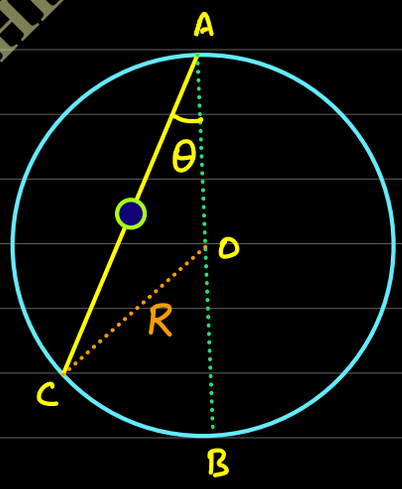
$\Rightarrow N = 600$ Newton.

Thus now we can say that the man weight 60Kg (Talking in mass equivalents).





EXAMPLE: A bead begins to slide on a frictionless thread tied to topmost point of a vertical circle as shown in the figure. If the bead was released from rest at point A, after how much time will it reach C?



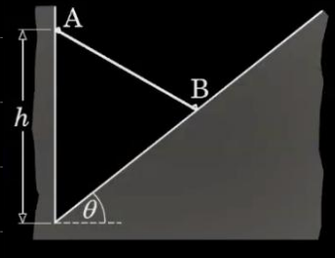
Solution: The bead accelerates along AC, with an acceleration $mg\cos\theta/m$ i.e $g\cos\theta$.

Now $AC = 2R\cos\theta$ ($\because \triangle OA$ is isosceles). Now using Kinematics of constant acceleration: $\frac{1}{2} g\cos\theta\, T^2 = 2R\cos\theta$

$\Rightarrow T = \sqrt{\frac{4R}{g}}$ Ans !!!

→ T is independent of θ, thus the time taken will be same along every chord. It only depends on radius of circle and g.

EXAMPLE: PATHFINDER: MCQ2:



In the setup shown a thread is taut between a nail A on a wall and a nail B on an inclined plane as shown in the figure. You can change the length of the thread by changing location of the nail B. A bead that can slide on the thread without friction is released from the nail A. What should the length of the thread be so that the bead reaches the nail B in shortest time?

(a) h (b) $h\cos\theta$

(c) $h\sec(0.5\theta)$ (d) $h\cos\theta\sec(0.5\theta)$

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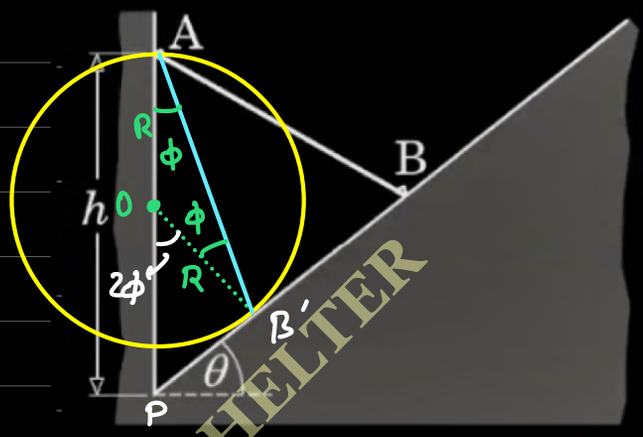
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Solution: Without loss of generality we can always imagine AB to be chord of a vertical circle whose topmost point is A. Thus our problem reduces to finding the vertical circle of smallest possible radius touching the incline, with B′ being the touching point.

Such a circle is shown in the adjacent figure. Now $h = R + R\sec 2\phi$

Also: $2\phi = \theta$ ($\because \Delta POB'$)

$\Rightarrow R = h/(1+\sec\theta)$

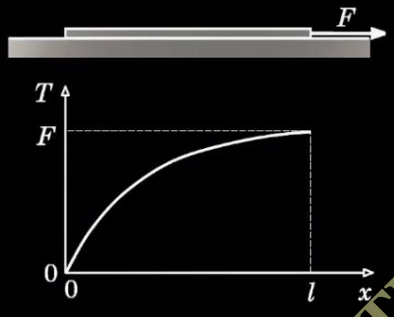


Now $l = 2R\cos\phi = \frac{2h}{1+\sec\theta}\cdot\cos(\theta/2)$

$= \frac{2h\cos\theta\cos(\theta/2)}{1+\cos\theta} = \frac{2h\cos\theta\cos(\theta/2)}{2\cos^2(\theta/2)} \Rightarrow$

$l = h\cos\theta\sec(\theta/2)$ Ans.

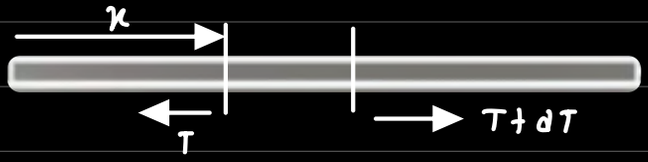
EXAMPLE: PATHFINDER MCQ3:



A rope of length l placed straight on a frictionless horizontal floor is pulled longitudinally by a force F from one of its ends. Tensile force T developed in the rope at a distance x from the rear end varies with x as shown in the graph. What can you certainly conclude about density of the rope?

(a) It is uniform.
(b) It decreases with distance x from the rear end.
(c) It increases with distance x from the rear end.
(d) It is maximum in the middle and decreases towards the ends.

Solution: Let a be the acceleration of the rope and $\lambda(x)$ be the mass per unit length of the rope. Now considering an element of length dx at a general coordinate x:



$(T+dT) - T = \lambda\, dx \times a$

$\Rightarrow \lambda = \frac{1}{a}\frac{dT}{dx}$ i.e. $\frac{1}{a}\times$ Slope of the graph. Thus we can conclude that the density of the rope decreases with increase in x. (option b).

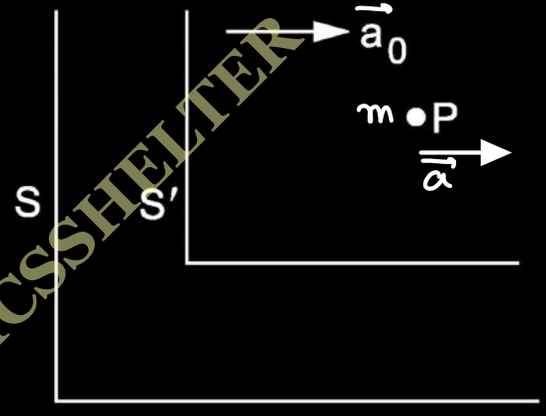
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PSEUDO FORCE: Recall that Newton's First Law of motion defines inertial reference frames (i.e frames in which Newton's Second Law is valid). For analysis in non-inertial frames we have to introduce fictitious forces to apply Newton's Second Law. In this chapter we will be dealing with translating non-inertial frames. We will start learning about rotating non-inertial frames when we study circular motion.

PSEUDO FORCE IN TRANSLATING FRAMES:

Let P be a particle of mass m moving with an acceleration $\vec{a}$ with respect to an inertial frame S. Now if $\vec{F}$ is the real force acting on it then:

$$\vec{F} = m\vec{a} \quad (1)$$



Now let S' be a non-inertial frame moving with an acceleration $\vec{a}_0$ wrt the inertial frame, then from such a frame, acceleration of P appears to be $\vec{a}' = \vec{a} - \vec{a}_0$. Now let $\vec{F}_p$ be the required pseudo force to make Newton's second law valid then:

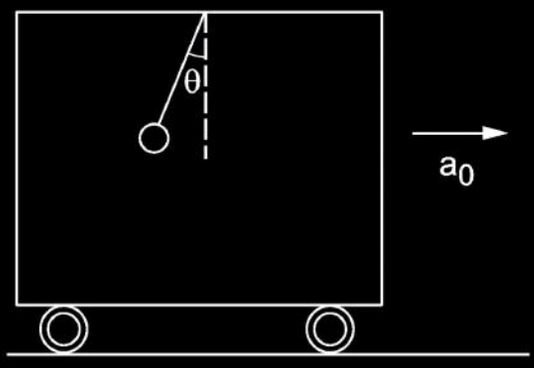
$$\vec{F}_p + \vec{F} = m(\vec{a} - \vec{a}_0) \quad (2)$$

From (1) & (2) : $\boxed{\vec{F}_p = -m\vec{a}_0}$

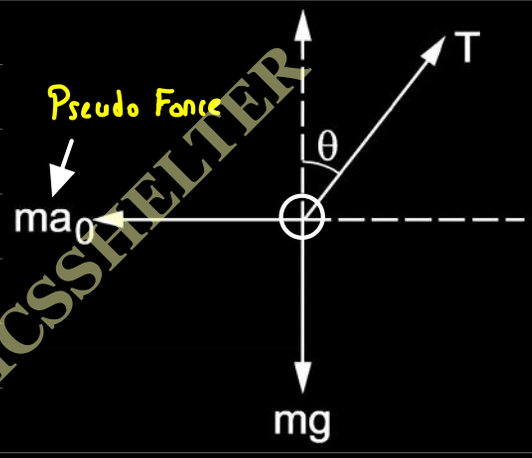
- Pseudo force acts opposite to the acceleration of Observation Frame
- Pseudo force does not depend on acceleration of observed object (or the real forces)
- Magnitude of pseudo force is mass of the <u>observed</u> object times the acceleration of <u>observation frame</u>.

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EXAMPLE: A pendulum is hanging from the ceiling of a car having an acceleration a_0 with respect to road. Find the angle made by the string with vertical.



Solution: The FBD of the pendulum from the car's frame is shown in the adjacent figure. (Carefully notice the magnitude and direction of pseudo force).



Now from car's frame the object appears to be in equilibrium thus we can use Lami's theorem:

$$\frac{T}{\sin 90^\circ} = \frac{mg}{\sin(90^\circ+\theta)} = \frac{ma_0}{\sin(180^\circ-\theta)}$$

This gives $\tan\theta = a_0/g$ & (Also $T = mg\sec\theta$).

USING GROUND FRAME: From ground frame, the bob appears accelerating with same acceleration as that of the car. Thus applying LOM in horizontal and vertical directions:

$$T\sin\theta = ma_0 \quad (1)$$

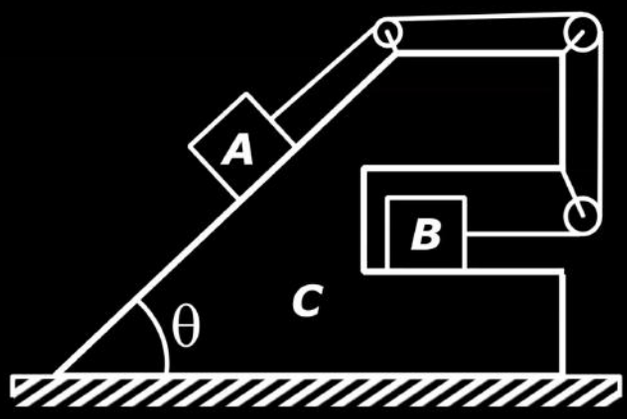
$$T\cos\theta = mg \quad (2)$$

$\Rightarrow \tan\theta = a_0/g$ and $T = mg\sec\theta$ &.

(We get the same answers again. The equations also remain the same essentially although our logic changed slightly. From one frame bob appears stationary while from the other frame, it appears to be accelerated).

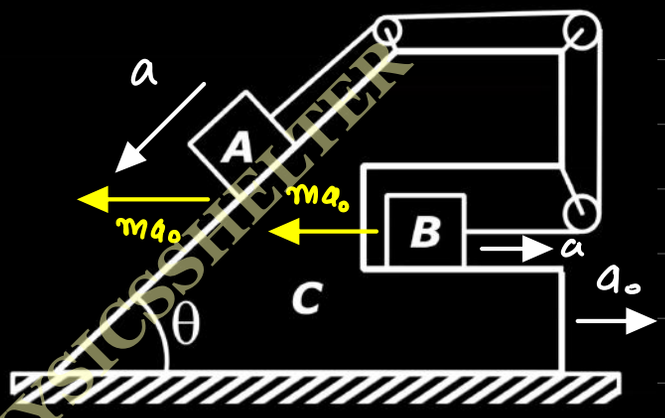
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Example: In the shown figure, each of the blocks a, b and c have mass m each and all surfaces are frictionless. The system is released from rest. Find
(a) Acceleration of C (a_0)
(b) Acceleration of A & B relative to C (a)
(c) Tension in the string.



Solution: For the system considered as a whole there is no external force in x direction. Thus $\Sigma m_i a_{ix} = 0$

Thus $m a_0 + m(a_0 + a) + m(a_0 - a\cos\theta) = 0$ ①



Now applying pulling force method from frame of C:

$$mg\sin\theta + ma_0\cos\theta - ma_0 = 2ma \quad ②$$

Now recognize that in equations ① & ② we have only two unknowns a & a_0. Thus solving for a and a_0:

$$a = \frac{3g\sin\theta}{5 + 2\cos\theta - \cos^2\theta}, \quad a_0 = \frac{g(-1+\cos\theta)\sin\theta}{5 + 2\cos\theta - \cos^2\theta}$$ Ans.

→ Think: Why is the acceleration of C (rightwards) (a_0 is negative).

Solution: With a_0 to the right momentum conservation will be clearly violated because relative to C, B must move rightwards for system to lose potential energy and magnitude of horizontal momentum of B in C frame is more than that of A.

Now we can easily calculate tension also, considering FBD of B:

$$T - ma_0 = ma \Rightarrow T = m(a_0 + a) = \frac{mg[2\sin\theta + \sin\theta\cos\theta]}{5 + 2\cos\theta - \cos^2\theta}$$ Ans.

Think: How to find Normal reaction from Ground:

Sol: $\Sigma F_{ext} = \Sigma m_i a_{iy} \Rightarrow 3mg - N = ma\sin\theta$!!!

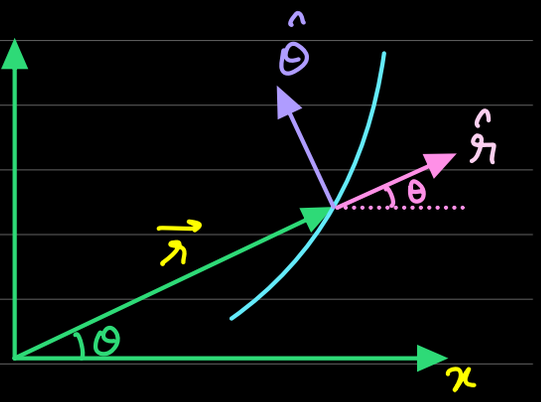
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Pseudo Force In Rotating Frame Seeing Only $\ddot{r}\hat{r}$ And $r\ddot{\theta}\hat{\theta}$:

Recall that we had derived an expression for acceleration of a particle is polar coordinates as:

$$\vec{a} = (\ddot{r} - \omega^2 r)\hat{r} + (r\alpha + 2\omega v_{rot})\hat{\theta} \quad (A)$$

and alternatively using vector cross products notation:



$$\boxed{\vec{a} = [\vec{\omega} \times (\vec{\omega} \times \vec{r})] + \ddot{r}\hat{r} + (\vec{\alpha} \times \vec{r}) + [2\vec{\omega} \times \vec{v}_{rot}]} \quad (1)$$

Now imagine a frame in which particle only appears to move away radially without any rotation (i.e a rotating frame which keeps its eye always toward the particle) in such a frame the acceleration just appears to be $\ddot{r}\hat{r}$. Now let pseudo force in such a frame be $\vec{F}_p$ and let $\vec{F}$ be the real force on the particle then to satisfy Newton's second law:

$\vec{F} + \vec{F}_p = m\ddot{r}\hat{r}$ but $\vec{F} = m\vec{a}$, thus: $\vec{F}_p = m\ddot{r}\hat{r} - m\vec{a}$ (2)

Now from equations (1) & (2):

$$\vec{F}_p = -m\,\vec{\omega} \times (\vec{\omega} \times \vec{r}) - m(\vec{\alpha} \times \vec{r}) - 2m\,\vec{\omega} \times \vec{v}_{rot} \quad (3)$$

Or if we used equation (A),

$$\vec{F}_p = m\omega^2 r\hat{r} - mr\alpha\hat{\theta} - 2m\omega v_{rot}\hat{\theta} \quad (4)$$

The three terms obtained in expression for pseudo force have specific names, described in following sections.

Centrifugal Force: The term $m\omega^2 r\hat{r}$ is called centrifugal force (vectorially $-m\,\vec{\omega} \times (\vec{\omega} \times \vec{r})$. This force acts radially outwards. Here $\vec{\omega}$ is the angular velocity of the rotating frame. Thus centrifugal force:

$$\boxed{F_c = m\omega^2\vec{r} = -m\,\vec{\omega} \times (\vec{\omega} \times \vec{r})}$$

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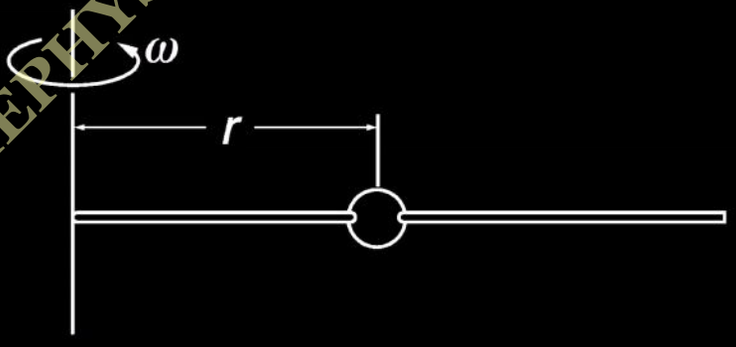
EULER FORCE: The term $-m\alpha r\hat{\theta}$ or $-m\vec{\alpha}\times\vec{r}$ is called the euler force. This acts only if our rotating frame has angular acceleration and acts in sense opposite to the sense of angular acceleration. Thus Euler Force:

$$\boxed{\vec{F}_E = -mr\alpha\hat{\theta} = -m\vec{\alpha}\times\vec{r}}$$

CORIOLIS FORCE: The term $-2m\vec{\omega}\times\vec{v}_{rot}$ is called coriolis force. This force acts only if in the rotating frame the particle appears to have some velocity and always acts $\perp$ to the $\vec{v}_{rot}$ (∵ it does no work and hence can't change kinetic energy). Thus, Coriolis force:

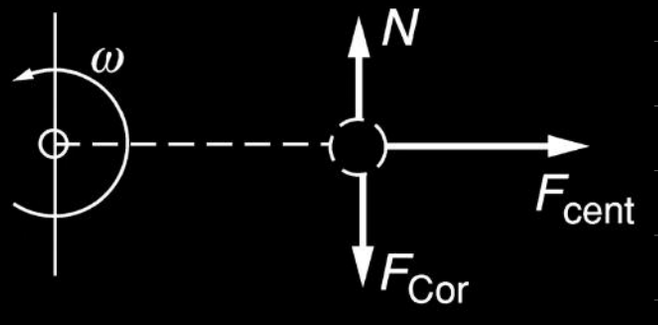
$$\boxed{\vec{F}_{cor} = -2m\vec{\omega}\times\vec{v}_{rot}}$$

EXAMPLE: A bead begins to slide on a horizontal frictionless rotating rod. The angular speed of rod is constant and equal to ω. Find (a) speed of the bead by the time it reaches a distance r from rotation axis assuming it started with negligible velocity from a point near the axis.



(b) Normal reaction from the bead on the rod when it is at the distance r. Neglect gravity.

Solution: (a) From rotating frame of the rod bead only has a radial outward motion. Now FBD of the bead from the rotating frame appears as shown in adjacent figure.



Now differential equation using LOM in radial direction:

$$m\ddot{r} = m\omega^2 r \Rightarrow mv\frac{dv}{dr} = m\omega^2 r$$

$$\Rightarrow \int_0^v v\,dv = \int_0^r \omega^2 r\,dr$$, where v is the speed as seen from rotating

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frame i.e V_{rot}. Thus $U_{rot} = \omega r$. ans.

(b) Now balancing forces perpendicular to the rod: $N = F_{cor}$

thus $N = 2m\omega V_{rot} = 2m\omega^2 r$, thus $N = 2m\omega^2 r$ ans.

Think: How much work is done on the system by the time bead reaches the distance r from axis.

Solution: From ground frame, the velocity is resultant of velocity as seen from rotating frame and velocity of the point of rotating frame just below the bead. Thus KE of bead from ground frame:

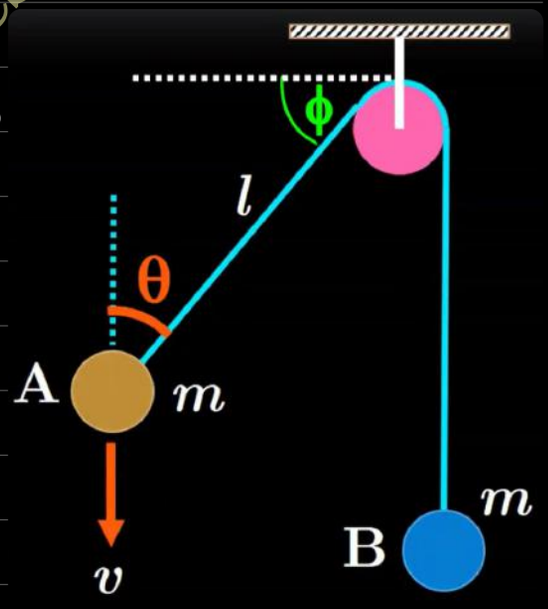
$KE = \frac{1}{2}m[(\omega r)^2 + (\omega r)^2] = m\omega^2 r^2$, this must be the work done on the system since this is change in KE.

EXAMPLE: In the shown system, bob A is given a downward velocity v and released. The pulley and string are massless and frictionless. The pulley may be considered to be a point.

(a) Draw FBD of A from rotating frame of the string centred at pulley

(b) Find acceleration of bob B using this

(c) Find angular acceleration of the rotating string.



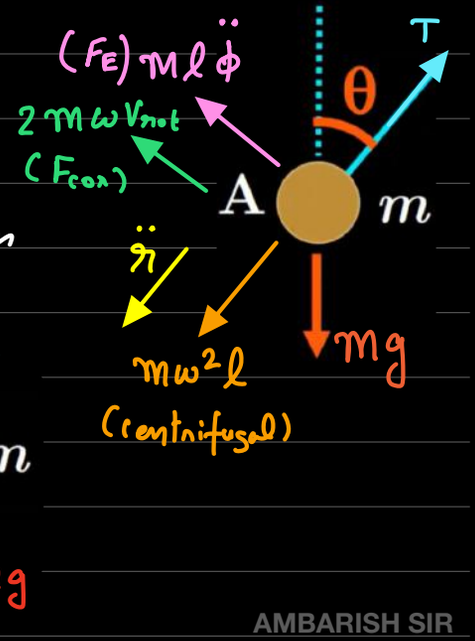
Solution: We choose a frame which has same angular velocity and angular acceleration as thread. From such a frame all pseudo and real forces are shown in adjacent FBD. From this frame the ball just has radial acceleration $\ddot{r}$. Thus writing NLM in radial direction:

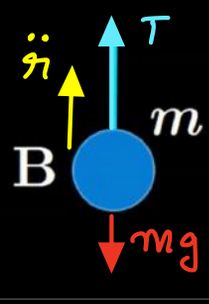
$mg\cos\theta + m\omega^2 l - T = m\ddot{r}$ ①

Now from FBD of ball B:

$T - mg = m\ddot{r}$ ②

Now solving ① & ②:





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$$mg(\cos\theta - 1) + m\omega^2 l = 2m\ddot{x} \Rightarrow \ddot{x} = \frac{1}{2}\left[\omega^2 l - g(1-\cos\theta)\right]$$

Now $\omega = v\sin\theta/l$ Thus: $\ddot{x} = \frac{1}{2l}\left[v^2\sin^2\theta - gl(1-\cos\theta)\right]$ Ans.

(This is acceleration of bob B).

Now balancing the forces $\perp$ to the string from rotating frame:

$$mg\sin\theta = ml\ddot{\phi} + 2m\omega v_{rot} \Rightarrow \ddot{\phi} = \frac{1}{l}\left[g\sin\theta - 2\omega v_{rot}\right]$$

Now $v_{rot} = v\cos\theta$, $\omega = v\sin\theta/l$ thus:

$$\alpha = \ddot{\phi} = \frac{1}{l}\left[g\sin\theta - 2\frac{v\sin\theta}{l}\cdot v\cos\theta\right]$$

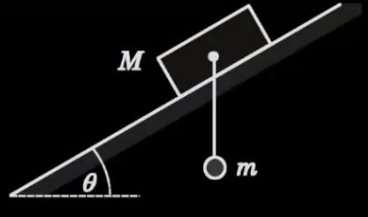
$$\Rightarrow \alpha = \frac{g\sin\theta}{l} - \frac{v^2\sin 2\theta}{l^2} \text{ Ans.}$$

→ If centre of rotating frame is also accelerating with an acceleration $\vec{a}$ then an additional pseudo force $-m\vec{a}$ will also have to be added. For instance suppose the pulley of previous example were fixed to ceiling of a car moving with an acceleration $\vec{a}$ along a road then a pseudo force $-m\vec{a}$ would be required on both bobs in addition to the other forces shown in the FBDs.

→ Home Task: Try to solve the previous example in ground frame. (Solution on channel: https://www.youtube.com/@ambarishphysics). (Search: Coriolis acceleration and Kinematics in general plane motion).

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EXAMPLE: PATHFINDER BUILD: 20:



A ball of mass m is suspended from a bar of mass M with a light inextensible cord. The bar can slide on a frictionless slope of inclination θ. Initially the bar is held motionless so that the ball also stays motionless as shown in the figure. Find accelerations of the bar and of the ball relative to the ground immediately after the bar is released. Acceleration due to gravity is g.

Solution: We will draw the FBD of the bob from the rotating frame centred on the string pivot (note that pivot is also accelerating).

AMBARISH SIR
https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

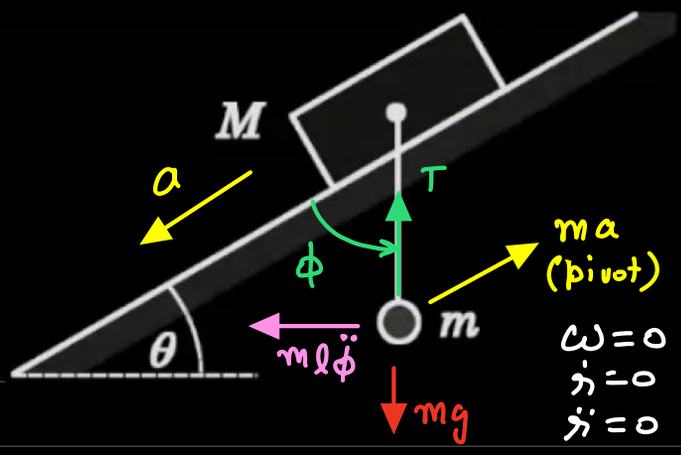
Now considering vertical equilibrium of the bob from our chosen frame,

$T + ma\sin\theta = mg$ ①

Now for block:

$T\sin\theta + Mg\sin\theta = Ma$ ②

Now from ① & ②,



$$a = \frac{(M+m)g\sin\theta}{M+m\sin^2\theta}$$ Ans.

(No coriolis, No centrifugal).

$$T = \frac{2Mmg\cos^2\theta}{M+m\sin^2\theta}$$, Now the real forces on m are only mg and T,

thus its acceleration from ground frame is simply, $a_B = (mg - T)/m$

thus, acceleration of bob is in vertical direction, given as:

$$a_B = g\left[1 - \frac{2M\cos^2\theta}{M+m\sin^2\theta}\right] = \frac{(M+m)g\sin^2\theta}{M+m\sin^2\theta}$$ Ans.

Alternatively, since velocity of pivot and bob are both zero, thus relative to pivot, bob has no vertical acceleration thus its vertical acceleration must be equal to that of pivot i.e $a\sin\theta$, and this must also be net acceleration, since there is no horizontal force.

Home task: Try this problem from ground frame.

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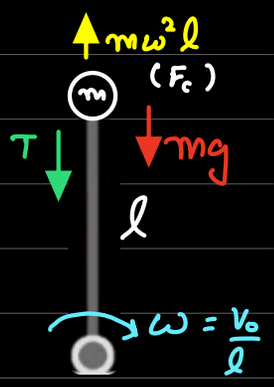
EXAMPLE: PATHFINDER: BUILD: 49:



A dumbbell is constructed by affixing small balls each of mass m at the ends of a light rod of length l. The dumbbell is held vertically with the lower ball resting on a horizontal floor as shown in the figure. The upper ball is released and given a horizontal velocity v_0 by a sharp hit. If the bottom ball immediately loses contact with the floor, what should the maximum length l of the dumbbell be? Acceleration due to gravity is g.

Solution: In the limiting case, the vertical acceleration of the lower ball may be assumed to be zero (Tension would just balance mg to make normal reaction zero).

Now we observe the upper ball from a rotating frame centred on lower ball and having a constant ω equal to that of the upper ball about the lower ball. Now in such a frame, $\ddot{x} = 0$



for the upper ball applying NLM:

$$m\omega^2 l = T + mg \quad (1)$$

No Coriolis

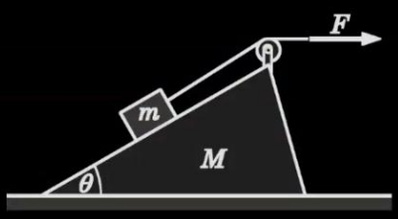
No Euler.

Now for lower ball, $T = mg$ (2) (∵ limiting case)
(∵ at the verge of lifting up).

Solving (1) & (2): $l = 2g/\omega^2 = 2g/(v_0/l)^2 = 2gl^2/v_0^2$

$$\Rightarrow \quad l = v_0^2/2g \quad \text{Ans.}$$

Example: Pathfinder: Build: 22:



A block of mass m placed on a wedge of mass M is attached at one end of a light inextensible cord that passes over an ideal pulley affixed on the top of the wedge as shown in the figure. The free end of the thread is pulled by a constant horizontal force F. If friction between the block and the wedge as well as between the wedge and the floor is absent, find acceleration of the wedge. Acceleration due to gravity is g.

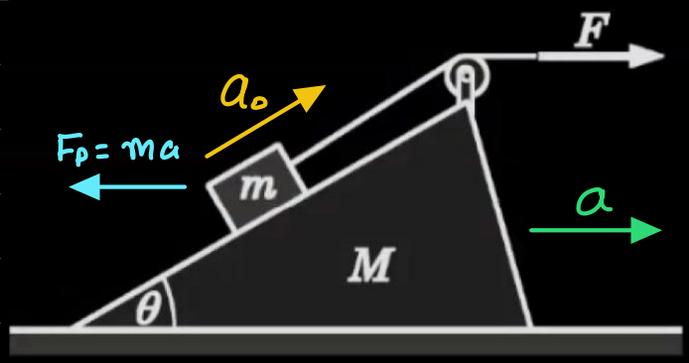
Solution: Considering combined FBD:

$$F = Ma + m(a + a_0\cos\theta) \quad (1)$$

Now using pulling force method for m from wedge frame:

$$F - mg\sin\theta - ma\cos\theta = ma_0 \quad (2)$$

Solving (1) & (2):



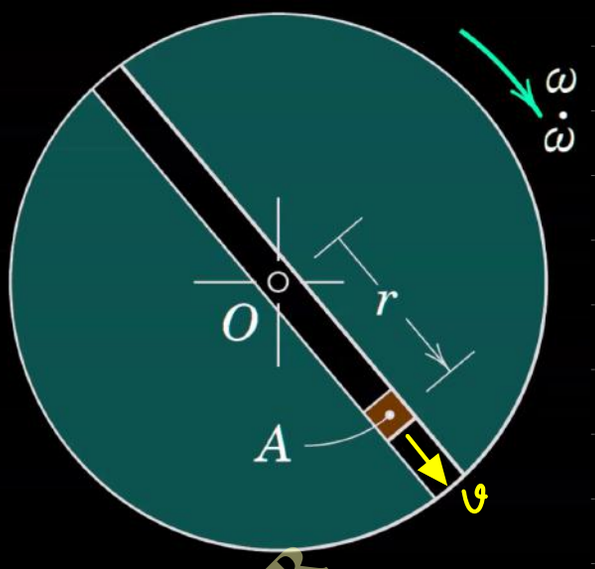
$$a = \frac{F(1-\cos\theta) + mg\sin\theta\cos\theta}{M + m\sin^2\theta}$$

Think: How will you find Normal Reaction from Ground?

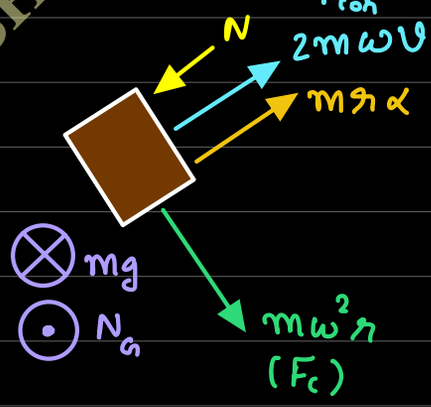
Solution: $\Sigma F_y = \Sigma m_i a_{iy}$. thus $N - (M+m)g = ma_0\sin\theta$ # N

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EXAMPLE: A turn table is rotating with an angular speed ω and angular acceleration $\alpha = \dot{\omega}$ as shown. At a certain instant, the block placed in its frictionless groove has a speed v relative to it. The mass of the block is m. (a) Draw the FBD of the block from the rotating frame (b) Calculate Normal reaction experienced by the block from side walls of the groove. (c) If α = Constant, constants, find the differential equation governing variation of r with time. Initially, $r=0$, $\omega_0 = 0$, $v_0 = 0$.



Solution: The adjacent figure shows the FBD of the block in top view as seen from rotating frame. (Particularly carefully notice the direction of coriolis force i.e. $-2m\vec{\omega} \times \vec{V}_{rel}$).



Now from groove frame acceleration is only radial ∴ tangential forces can be balanced.

Thus:

$N = 2m\omega v + mr\omega^2$ Ans (b). (Reaction from side wall).

Now: $\alpha = \frac{d\omega}{dt} \Rightarrow \omega = \alpha t + \text{Const}^{tt} \Rightarrow \omega = \omega_0 + \alpha t$ ($\because$ @ $t=0$, $\omega = \omega_0$).

Also: $m\omega^2 r = m\frac{d^2r}{dt^2} \Rightarrow m(\omega_0 + \alpha t)^2 r = m\frac{d^2r}{dt^2}$

EXAMPLE: For the previous problem assuming $\alpha = 0$, $\omega = \omega_0$ find $v(r)$ assuming $v(0) = 0$ and the block started from centre of turn table.

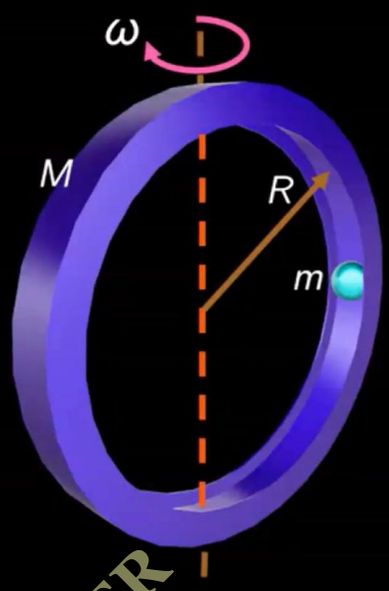
Solution: $m\omega^2 r = mv\frac{dv}{dr} \Rightarrow v\,dv = \omega^2 r\,dr \Rightarrow \frac{v^2}{2} = \frac{\omega^2 r^2}{2} + C$

but $C = 0$ $\because$ @ $r=0$, $v=0$ $\Rightarrow$ $v = \omega r$ Ans.

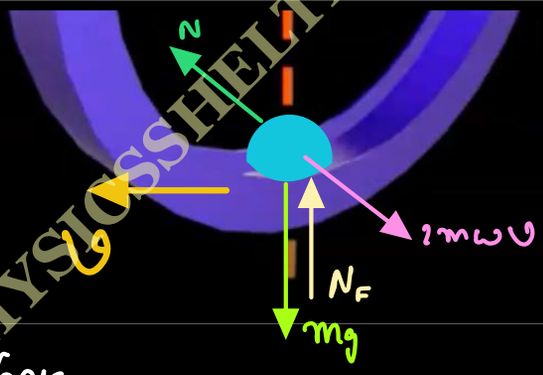
(From ground frame speed must be $\sqrt{2}\omega r$ since tangential speed is also ωr).

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Example: A ring rotating with angular velocity ω has a smooth groove in which a small particle was released. When the particle reached bottom most point, its speed relative to the groove is v. Find the normal reaction on the particles from side walls of the groove in this position. Angular velocity of the ring is maintained constant.



Solution: The adjacent figure shows the FBD of the particle at the bottommost position as seen from the rotating frame. From the frame of the groove, the particle has no sideways acceleration (just a centripetal acceleration v^2/R). Here there is no centrifugal or euler force because distance from rotation axis is zero and $\alpha = 0$). Now balancing sideways forces, $N = 2m\omega v$ Ans.



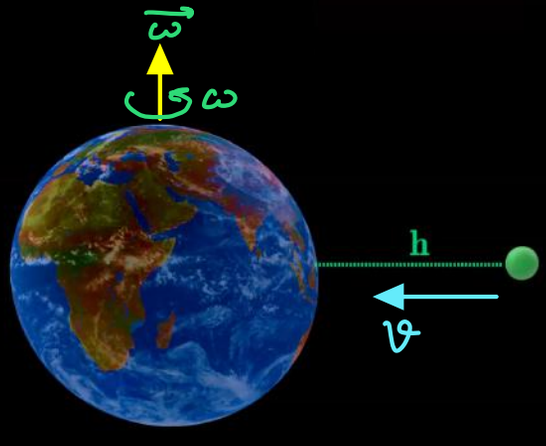
Think: What is the reaction from the floor?

Solution: $N_F - mg = mv^2/R$ ($\because$ (centripetal acc^n is v^2/R)

$\Rightarrow N_F = mg + mv^2/R.$

Example: Irodov 1.117:

At the equator a stationary (relative to the Earth) body falls down from the height $h = 500$ m. Assuming the air drag to be negligible, find how much off the vertical, and in what direction, the body will deviate when it hits the ground.



Solution: As the stone falls it experiences a sideways shift due to coriolis force given by $-2m\vec{\omega} \times \vec{v}$. Now $\vec{\omega}$ is directed Northwards and $\vec{v}$ is directed downwards thus coriolis force is directed eastwards

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therefore the stone will experience a slight eastward drift ($\because \omega$ is very small). Now $v = gt$ and $T = \sqrt{2h/g}$. Thus coriolis acceleration:

$$a_{cor} = 2m\omega gt/m = 2\omega gt \quad \text{(eastwards)}.$$

Thus eastward velocity at any general time, $\int_0^t a_{cor}\,dt = \omega g t^2$

Thus $\frac{dx}{dt} = \omega g t^2 \Rightarrow \int_0^x dx = \int_0^{\sqrt{2h/g}} \omega g t^2 dt = \frac{\omega g}{3}\left[\frac{2h}{g}\right]^{3/2}$

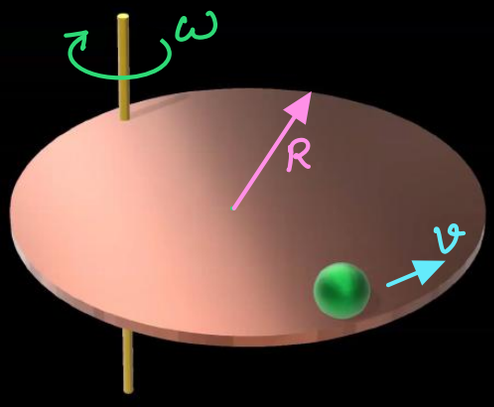
$\Rightarrow x = \frac{2}{3}\omega h\sqrt{\frac{2h}{g}} \approx 24\text{cm}$ Ans.

→ By the same logic, if a stone were projected vertically upwards, it will have a westward drift.

→ Due to coriolis forces, water level at the two banks may vary slightly.

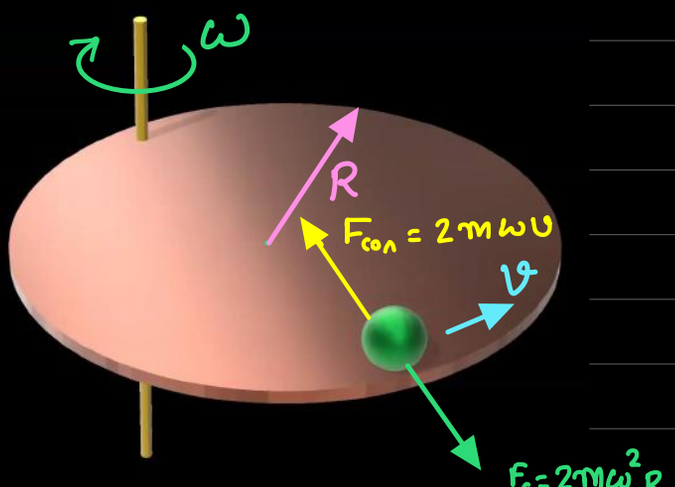
EXAMPLE: IRODOV 1.114:

A horizontal disc of radius R rotates with a constant angular velocity ω about a stationary vertical axis passing through its edge. Along the circumference of the disc a particle of mass m moves with a velocity that is constant relative to the disc. At the moment when the particle is at the maximum distance from the rotation axis, the resultant of the inertial forces F_{in} acting on the particle in the reference frame fixed to the disc turns into zero. Find:
(a) the acceleration w' of the particle relative to the disc;
(b) the dependence of F_{in} on the distance from the rotation axis.



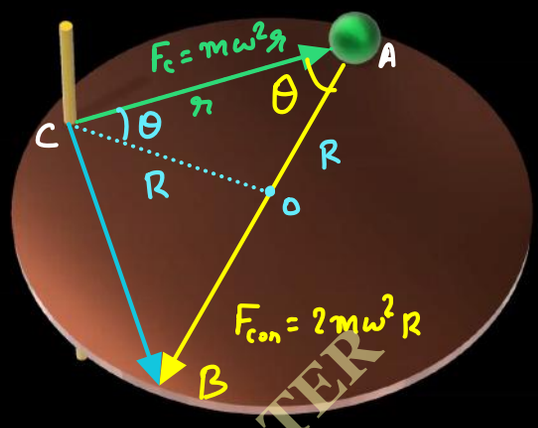
Solution: At the diametrically opposite point, according to the question, pseudo forces cancel each other. Thus coriolis force must cancel the centrifugal force. Thus:

$$2m\omega^2 R = 2m\omega v \Rightarrow$$
$$v = \omega R \text{ (proved)}.$$
$$\Rightarrow w' = \frac{v^2}{R} = \omega^2 R \quad (\because \text{centripetal acc}^n)$$
Ans



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Now coriolis force is always perpendicular to v_{rot} thus it is always directed towards centre of the disc and has a magnitude $2m\omega v$ i.e $2m\omega^2 R$, while centrifugal force is always $m\omega^2 r$. Now Consider triangle CAB, we see that centrifugal force is along CA and proportional to r and coriolis force is along AB and proportional to 2R i.e diameter of the circle, with proportionality constants for both forces being $m\omega^2$. Thus by similarity and triangle law of addition, the resultant must be along CB and with same proportionality constant. Also angle in a semicircle is 90°



thus $CB = 2R \sin\theta = 2R\sqrt{1-\cos^2\theta} = 2R\sqrt{1-r^2/4R^2} =$

Thus resultant inertial force : $F_{in} = m\omega^2 \times 2R\sqrt{1-(r/2R)^2}$

$\Rightarrow F_{in} = m\omega^2\sqrt{4R^2-r^2} = m\omega^2 r\sqrt{\left(\frac{2R}{r}\right)^2-1}$ Ans.

Think: What is the real force magnitude and direction?

Solution: Net resultant force must act along OB since from disc frame particle has net accelerating $m\omega^2\vec{OB}$, thus:

$$m\omega^2\vec{CB} + \vec{F}_{real} = m\omega^2\vec{OB} \Rightarrow \vec{F}_{real} = m\omega^2(\vec{OB} - \vec{CB})$$

i.e $m\omega^2\vec{OC}$, thus real force is always directed parallel to $\vec{OC}$ and has a magnitude $m\omega^2 R$.

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